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Abstract

We provide an analytically tractable theoretical model for the optimal supply of reserves and the design of standing liquidity facilities under uncertainty about banks' demand for reserves. Absent a lending facility, uncertainty creates a precautionary motive to supply additional reserves. Introducing a lending facility generates a tradeoff between rate control and market disintermediation. If the facility rate is set optimally, the presence of the facility reduces the optimal supply of reserves. If the central bank cannot set the lending rate optimally, the optimal supply of reserves can be higher or lower than absent a facility.

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This paper presents preliminary findings and is being distributed to economists and other interested readers solely to stimulate discussion and elicit comments. The views expressed in this paper are those of the author(s) and do not necessarily reflect the position of the Federal Reserve Banks of New York and San Francisco, the Federal Open Market Committee, or the Federal Reserve System. Any errors or omissions are the responsibility of the author(s).

To view the authors' disclosure statements, visit https://www.newyorkfed.org/research/staff_reports/sr1077.html.

1 Introduction

The ways in which central banks supply reserves and control interest rates in alignment with their monetary policy goals have evolved over the past two decades. Prior to the global financial crisis (GFC), many central banks opted for implementation frameworks whereby the quantity of reserves in excess of regulatory requirements is small, reserve supply intersects banks' demand in the steeply-sloped region of the demand curve, and control over short-term interest rates is exercised through regular open market operations. In response to the GFC and again following the COVID-19 pandemic, many central banks expanded their balance sheets through quantitative easing programs funded in part by large-scale increases of central bank reserves. These developments necessitated fundamental changes in how central banks approach the provision of reserves while maintaining control of short-term rates.

As a result of their experiences in managing large balance sheets, several central banks—including the Federal Reserve, the European Central Bank, and the Bank of England—have significantly modified their strategies for supplying reserves and controlling interest rates. Although the approaches differ in important ways, a common feature of these approaches is that they combine the provision of a quantity of interest-bearing reserves with a standing lending facility designed to limit upward movements in short-term rates.

This paper analyzes the optimal supply of reserves using an analytically tractable model of the supply and demand of reserves that has applicability to a range of institutional structures and policy objectives. The key building block of this analysis is a nonlinear demand curve for central bank reserves that is subject to uncertainty, consistent with the theoretical and empirical evidence (Poole, 1968; Afonso and Lagos, 2015; Bianchi and Bigio, 2022; d'Avernas et al., 2023; Lagos and Navarro, 2023; Afonso et al., 2025; and

Copeland et al., 2025). The analytical model is simple and flexible, making it a useful framework to examine the effects of the variety of circumstances that underlie central banks' decision-making regarding the supply of reserves and the design of liquidity facilities.

The reserve demand curve represents the downward relationship between the price of reserves—measured as the spread between the short-term interest rate in the market for reserves and the interest rate paid on reserve balances held at the central bank—and the aggregate quantity of reserves. This demand is the result of a tradeoff: banks receive benefits from holding reserves but also incur costs. Since the net benefits decline with the quantity of reserves, the curve is nonlinear and has a lower asymptote. If a lending facility is present, the curve has also an upper asymptote because the facility sets a ceiling on market rates.

The central bank controls two policy instruments: it sets the ex-ante aggregate level of reserves available to the banking system and the rate at which banks can borrow from the standing lending facility (for any given level of the interest rate on reserve balances). In our model, these decisions are made in advance of the realization of uncertainty about the demand for reserves. The analysis is static in nature and abstracts from learning, transition dynamics, and decisions regarding the assets held by the central bank.

The central bank sets its two policy instruments to minimize a linear combination of the expected losses associated with three different aspects of supplying reserves to the banking system. First, the central bank aims to minimize deviations of the policy rate (i.e., the interest rate in the market for reserves) from its target level. Second, the central bank may have a target for the quantity of reserves that reflects goals that are unrelated to the control of short-term rates, such as the functioning of financial markets and financial stability. Third, usage of the standing lending facility may be associated with non-pecuniary costs for the central bank, such as disintermediation of the interbank

market, which hinders the informativeness of prices and market efficiency.

This analysis yields four novel insights regarding the optimal ex-ante supply of reserves and the design of standing liquidity facilities. First, we fully characterize the optimal supply of central bank reserves and the optimal lending facility spread (with respect to the interest rate on reserve balances), along with the associated market equilibrium, in the presence of uncertainty about the demand for reserves. Second, we show that, absent a lending facility, the optimal ex-ante supply of reserves under uncertainty is greater than it would be absent uncertainty, while the average interbank market spread may be higher or lower than in the deterministic setting. Third, in the presence of a lending facility, the optimal facility spread increases with the penalty on facility usage in the central bank's objective function; in the limiting case of no penalty, the optimal lending facility spread equals the target market spread. Fourth, given an optimal choice of the lending facility spread, the optimal ex-ante supply of reserves is smaller than it would be absent a lending facility; if, on the other hand, the lending facility rate is not set optimally, the optimal ex-ante level of reserves may be lower or higher than it would be without the facility. Taken together, these results show that the optimal choices for the supply of reserves and the design of liquidity facilities depend critically on institutional factors, such as market frictions or constraints related to liquidity facilities, as well as central bank preferences over the various goals related to the supply of reserves.

This paper makes two main contributions to the literature and policy debate. First, by deriving the optimal supply of central bank reserves under uncertainty, it adds to the academic literature on the optimal supply of money going back to Friedman (1969). Relevant related research includes Afonso et al. (2021), Yang (2022), d'Avernas et al. (2023), Lagos and Navarro (2023), Afonso et al. (2025), and Copeland et al. (2025), who empirically study how reserve scarcity led to severe dislocations of U.S. money market rates in September 2019. Other recent research related to this paper includes Martin

et al. (2019), Bigio and Sannikov (2021), and Bianchi and Bigio (2022), who study the problem of optimal monetary policy implementation from the perspective of monetary policy transmission to the real economy. Second, this paper contributes to recent policy debates about the optimal framework for monetary policy implementation by providing a general, analytically tractable, theoretical model of monetary policy implementation that includes the fundamental tools used by central banks to control money-market rates, encompassing the different systems adopted across jurisdictions. This model can therefore be used to formally analyze the trade-offs that affect the design of central banks' operational frameworks.

The paper is organized as follows. Section 2 describes the institutional background, the theoretical model of the reserve demand curve, how a standing lending facility affects the curve, and the central bank optimization problem. Section 3 derives the optimal ex-ante supply of reserves without a lending facility. Section 4 derives the optimal design of the lending facility and the implications for the optimal ex-ante supply of reserves. Section 5 discusses rate control in different implementation frameworks and the trade-offs central banks face in the design of their frameworks. Section 6 concludes.

2 Monetary Policy Implementation and Reserves Demand

Central banks rely on two main sets of tools to implement monetary policy: instruments that guide the policy rate by adjusting the quantity of reserves in the banking system and instruments that steer the policy rate by changing the rates controlled by the central bank. The choice of toolkit depends on the sensitivity of the policy rate to changes in the supply and demand for reserves, which itself varies with the quantity of reserves in the system.

2.1 Institutional Background

The Federal Reserve has implemented monetary policy with different frameworks over time. Before the GFC of 2007–09, reserves were scarce, fluctuating within the steeply sloped region of the reserve demand curve. In this regime, control of the policy rate—the federal funds rate—was maintained through daily changes in the supply of reserves. Every day, the Federal Reserve would forecast banks’ demand for reserves and those factors that affect the supply outside its control; given these forecasts, it would then inject or drain reserves through daily repurchase agreement (repo) operations with primary dealers to ensure that the policy rate remained close to its target. Currently, several jurisdictions—including the central banks of Korea, Turkey, and Sweden—continue to carry out monetary policy in this way.

With the onset of the GFC, the Federal Reserve undertook several rounds of quantitative easing, purchasing longer-term financial assets to improve conditions in financial markets and stimulate the economy (Bernanke, 2020). As a result of these purchases, reserves in the banking system increased substantially, moving into the lower asymptote of the reserve demand curve, and the policy rate became insensitive to small adjustments in the supply.¹ For this reason, the Federal Reserve transitioned to implementing monetary policy primarily through administered rates. In this new regime, rather than adding or subtracting reserves daily, the Federal Reserve adjusts the interest rates of its deposit and lending facilities to maintain the policy rate within the target range. The European Central Bank (ECB), Bank of England, and Bank of Canada had similar experiences.

Since then, several central banks have reviewed and modified their frameworks. In January 2019, the Federal Reserve completed its review and announced that it will continue to implement monetary policy in a regime with an ample supply of reserves where rate

¹Reserves increased from 0.6% of bank assets during 2000:Q1–2008:Q2 to 12.5% during 2008:Q3–2020:Q1. As a result of the quantitative easing implemented by the Federal Reserve in response to the COVID-19 pandemic, this ratio increased further to 15.1% during 2020:Q2–2025:Q2.

control is exercised primarily through the setting of its administered rates. To strengthen rate control, the Federal Reserve established a standing repo facility (SRF) in 2021 to set a ceiling on short-term rates and thus limit occasional upward pressure.²

In 2024, the ECB concluded the review of its framework and also decided to continue to implement monetary policy through administered rates. Among other changes, the ECB narrowed the spread between the rate at the lending facility for the main refinancing operations and the rate at the deposit facility (European Central Bank Governing Council (2024)). Similarly, the 2022–24 Bank of England’s review also resulted in a decision to control the policy rate through administered rates and reliance on a standing lending facility, the short-term repo facility, to provide unlimited amounts of reserves to the banking system (Bank of England (2022)). The novelty of the Bank of England approach is that it sets the spread between the upper and lower bounds of the policy rate to zero, aiming to make the policy rate constant and independent of aggregate reserves and to incentivize banks to borrow from the central bank as much as they need (Bank of England (2024)). Other central banks have also reviewed and modified their implementation frameworks recently, including the Bank of Canada (Bank of Canada (2025)) and the Reserve Bank of Australia (Kent (2025)).

2.2 A Model of Reserve Demand

The reserve demand curve represents the relationship between the interest rate at which banks are willing to trade reserve balances with each other and the aggregate reserves in the banking system. Economic theory predicts that this curve is nonlinear (Poole (1968)): it decreases with reserves and levels out with higher reserve balances. The intuition is simple: banks hold reserves to meet their liquidity needs (e.g., payments, regulatory

²The Federal Reserve also established a repo facility for foreign and international monetary authorities (FIMA repo facility) to address pressures in global dollar funding markets.

requirements). As the quantity of reserves increases, the net benefit of an additional unit of reserves declines, and the relation between price and quantity becomes flatter. For low levels of reserves, the slope of the curve is steep, and a small change in the quantity of reserves results in a significant change in the market rate; this region of the demand curve is called the “scarce reserves” region. At higher levels of reserves, the slope is still negative but smaller in magnitude; this is often referred to as the “ample reserves” region. At sufficiently high levels of reserves, the curve is flat, i.e., bank demand is satiated; this is called the “abundant reserves” region.³

In practice, it is convenient to write the demand curve in terms of the spread between the rate at which banks trade reserves and the rate at which the central bank remunerates reserve balances. In this way, the demand curve is unaffected by changes in banks’ opportunity cost of lending reserves driven by the stance of monetary policy.⁴ We start by analyzing a simplified version of the model in which there are only two regions of demand for reserves: ample and abundant reserves; in Section 3.4, we present an extension of the model with a third region with scarce reserves.

We model the spread between the interbank market rate and the interest rate on reserve balances, S , as a piecewise-linear function of aggregate reserves, X :

$$S = \begin{cases} \bar{S} + \nu - \alpha(X - \bar{X} - \epsilon) & \text{if } X < \bar{X} + \epsilon, \\ \bar{S} + \nu & \text{else,} \end{cases} \quad (1)$$

where $\bar{X} + \epsilon$ denotes the level of reserves above which the demand for reserves becomes flat (abundant reserves), $\alpha > 0$ is the steepness of the demand curve if $X < \bar{X} + \epsilon$ (ample reserves), and $\bar{S} + \nu$ is the spread if reserves are abundant (floor of the demand curve).

³See Ennis and Keister (2008), Afonso and Lagos (2015), Armenter and Lester (2017), Afonso et al. (2019), Kim et al. (2020), Bigio and Sannikov (2021), Bianchi and Bigio (2022), and Lagos and Navarro (2023).

⁴This representation of the demand curve has been broadly adopted by recent theoretical and empirical papers. See Bigio and Sannikov (2021); Bianchi and Bigio (2022); Lopez-Salido and Vissing-Jorgensen (2022); Lagos and Navarro (2023); Acharya et al. (2023); Afonso et al. (2025).

Consistent with the theory, the slope of the demand curve in the abundant reserves region is zero.⁵ We assume that $\bar{X}$, α , and $\bar{S}$ are known to the central bank, whereas ϵ and ν are uncertain. In Section 3.4, we present an extension of the model that also includes uncertainty about the slope of the demand curve.

Absent uncertainty, the demand curve is $S = \max(\bar{S}, \bar{S} - \alpha(X - \bar{X}))$, which is shown by the solid black lines in Figure 1. In the flat region of the reserve demand curve, where reserves are abundant, the interbank market rate is anchored to the interest rate on reserve balances. In the absence of frictions, the spread between these rates, $\bar{S}$, is zero. In the presence of frictions such as balance-sheet costs or market segmentation, however, the spread can be negative (Afonso et al., 2019; Kim et al., 2020).

Figure 1 also shows how the two elements of uncertainty in our model affect the demand for reserves. Each type of uncertainty is represented by a mean-zero shock with finite variance that is uncorrelated with the other shock. The shock ϵ represents uncertainty about the location of the kink in the demand curve; it is assumed to have a continuous probability density function $g(\cdot)$, a cumulative distribution function $G(\cdot)$, and variance σ_ϵ^2 . This horizontal shock can be interpreted as an unobserved horizontal shift to the demand curve due to changes in banks' demand for liquidity; it can also be interpreted as a shock to the supply of reserves caused by factors outside the control of the central bank.⁶ The shock ν , with variance σ_ν^2 , reflects uncertainty about the level of the floor of the demand curve and affects the spread equally along the curve. This shock can be interpreted as an unobserved vertical shift to the demand curve due to changes in the frictions that determine $\bar{S}$.

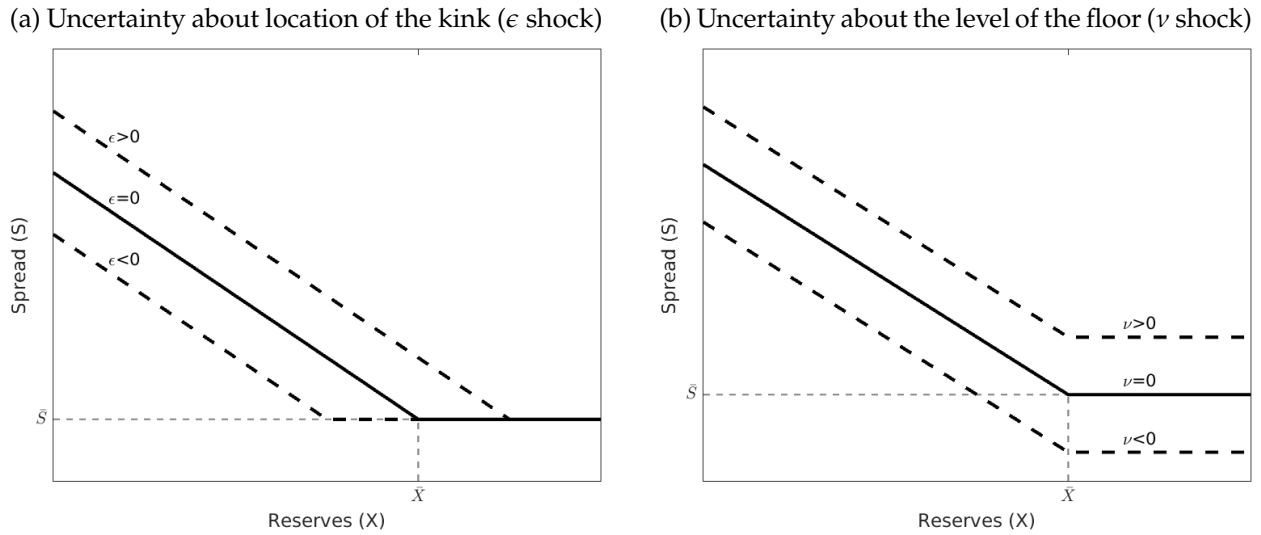
Both of our key modeling assumptions—a nonlinear reserve demand curve with a

⁵Our analysis and key results carry over if the slope in the abundant reserves region is negative, as long as it is smaller in magnitude than the average slope in the ample reserves region.

⁶In many jurisdictions, currency in circulation (banknotes and coins) and the government account at the central bank (e.g., the Treasury General Account in the U.S.) constitute important factors outside the control of the central bank that affect the level of reserves in the banking system.

lower flat region and the presence of horizontal and vertical shifts—are consistent with the empirical evidence about banks’ demand for reserves in the United States (Afonso et al. (2025)); see Appendix C for more details.

Figure 1: A Graphical Representation of Uncertainty about the Demand for Reserves. These figures show the effects of the two types of uncertainty on the demand for reserves in equation (1). Panel (a) shows the effect of uncertainty about the horizontal location of the kink $\bar{X}$ (ϵ shock); panel (b) shows the effect of uncertainty about vertical location (the floor) of the demand curve $\bar{S}$ (ν shock).



2.3 Standing Lending Facilities

Central bank standing lending facilities enhance rate control by establishing a ceiling on the rates at which banks are willing to borrow from each other: absent frictions, if a lending facility is available, banks should not borrow in the market at rates above the facility rate.⁷ Central banks design their lending facilities weighting the benefits and costs of public liquidity provision: they aim to balance the advantages for rate control with concerns about market disintermediation, which could hinder monitoring and price discovery.⁸

⁷Frictions such as stigma and balance-sheet costs may push the upper bound higher than the facility rate.

⁸See Federal Reserve (2021) for considerations on the establishment of standing repo facilities in the U.S.

In our model, we assume the implementation framework includes a fixed-rate lending facility. We denote the spread between the lending facility rate and the interest rate on reserve balances by $\bar{S}_L$. We begin our analysis by assuming that the facility spread is exogenous, that is, the central bank chooses the ex-ante reserve supply taking the facility spread as given; this could be the case if legal or other institutional constraints restrict the setting of the lending rate. We then relax this assumption and allow the central bank to jointly choose the ex-ante supply of reserves and the lending rate spread.

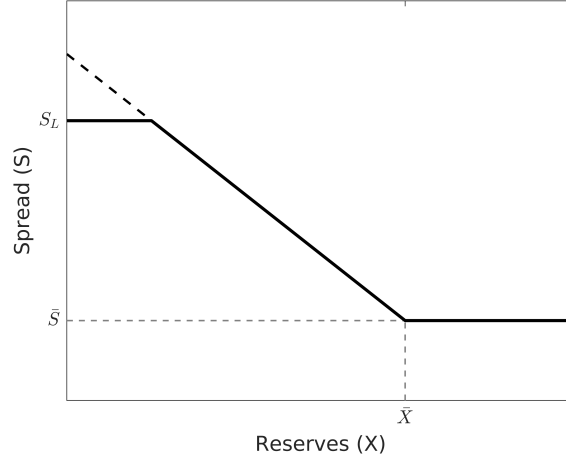
We abstract from stigma and other frictions and assume that the facility spread, $\bar{S}_L$, sets a ceiling on the market spread, as shown in Figure 2. We also assume that banks can borrow from the facility after uncertainty is realized. When banks' demand for reserves intersects the total ex-ante supply ($X + \epsilon$) at a spread above $\bar{S}_L$, banks borrow from the facility the amount of reserves L that makes the total supply (ex-ante supply plus borrowed reserves) equal to the level of reserves corresponding to $\bar{S}_L$ on the demand curve. This quantity is a random variable, described by:

$$L = \begin{cases} -\frac{1}{\alpha} (\bar{S}_L - \bar{S} - \nu) - (X - \bar{X} - \epsilon) & \text{if } \bar{S} + \nu - \alpha(X - \bar{X} - \epsilon) > \bar{S}_L, \\ 0 & \text{else,} \end{cases}$$

where $\bar{\epsilon}_L = X - \bar{X} + \frac{\bar{S}_L - \bar{S}}{\alpha}$ is the cutoff value for the demand shock at which the facility is first utilized; then, the probability of using the lending facility is $1 - G(\bar{\epsilon}_L)$, and the demand for reserves is given by

$$S = \begin{cases} \bar{S}_L & \text{if } X < \bar{X} + \epsilon - \frac{1}{\alpha} (\bar{S}_L - \bar{S} - \nu), \\ \bar{S} + \nu - \alpha(X - \bar{X} - \epsilon) & \text{if } \bar{X} + \epsilon - \frac{1}{\alpha} (\bar{S}_L - \bar{S} - \nu) < X < \bar{X} + \epsilon, \\ \bar{S} + \nu & \text{else.} \end{cases} \quad (2)$$

Figure 2: **Reserve Demand Curve with a Standing Lending Facility.** This figure shows the effect of a fix-rate lending facility on the demand for reserves.



2.4 The Central Bank Optimization Problem

The central bank has two main tools to steer the policy rate towards its target. First, it can alter the ex-ante supply of reserves X , through outright asset purchases (or sales) or repo operations. Second, if a lending facility has been established, the central bank can change the spread between the facility rate and the interest rate on reserve balances $\bar{S}_L$, to adjust the ceiling on market rates.

Given the demand curve in equation (2), the central bank uses these two implementation tools to minimize the following quadratic objective function

$$\begin{aligned} \mathcal{L} &= \min_{X, \bar{S}_L} \left\{ \mathbb{E} \left[\frac{1}{2} \left((S - \hat{S})^2 + \lambda (X - \hat{X})^2 \right) + \mu L \right] \right\} \\ &= \min_{X, \bar{S}_L} \left\{ \frac{1}{2} \left((\mathbb{E}S - \hat{S})^2 + \text{Var}[S] + \lambda (X - \hat{X})^2 \right) + \mu L \right\}, \end{aligned} \quad (3)$$

where $\hat{S}$ and $\hat{X}$ are the target spread and the target level of reserves that the central bank aims to achieve; L is the amount that banks borrow from the lending facility; $\lambda \geq 0$ is the relative weight given to deviations from the reserves target $\hat{X}$; and $\mu \geq 0$ is the penalty related to facility usage.

Note that in the central bank loss function (3), we treat the ex-ante supply of reserves, X , differently from the reserves borrowed from the facility, L . This specification is consistent with the theoretical arguments in Section 5, which suggest that the ex-ante supply of reserves and reserves borrowed from a lending facility have different costs and benefits for the central bank. Also note that the central bank optimization problem is static; that is, we assume that the degree of uncertainty is fixed and there is no ability to learn.

The target spread, $\hat{S}$, reflects efficiency and welfare considerations that are related to the level of money-market spreads in the economy. A natural benchmark is the Friedman rule: absent frictions, to maximize economic efficiency, it would be optimal not to set a spread between the rate in the market for reserves and the interest rate on reserve balances ($\hat{S} = \bar{S} = 0$). However, in the presence of frictions, such as bank balance-sheet costs or household credit constraints, it can be optimal for the central bank to target a spread above the lower asymptote of the reserve demand curve, $\hat{S} > \bar{S}$ (Bigio and Sannikov, 2021).

The central bank may also have a separate target for the level of reserves, $\hat{X}$. This target reflects considerations about the quantity of reserves in the system that are not directly related to the level of money-market spreads in the economy; typical examples are the political economy concerns about a large central bank balance-sheet (Bindseil, 2016) and the concerns about the possible disintermediation of the banking system and its impact on market discipline and price discovery (Borio, 2023). In Section 5.2, we discuss these considerations in more detail. If these concerns are sufficiently large, the central bank may target a level of reserves below the region of abundant reserves ($\hat{X} < \bar{X}$).

3 Optimal Policy Without a Lending Facility

In this section, we analytically derive the optimal supply of reserves by the central bank given the demand curve in equation (1), i.e., in the absence of a lending facility. In this

setting, the central bank has only one instrument: the ex-ante supply of reserves X .

3.1 Optimal Supply of Reserves Absent Uncertainty

We first analyze the optimal supply of reserves in the deterministic version of the model that excludes the two shocks ϵ and ν in equation (1). The proofs for the results in this section are in Appendix A.1.

We begin by assuming that both central bank targets are in the ample region ($\hat{X} > \bar{X}$ and $\hat{S} > \bar{S}$). Taking the first-order condition for problem (3) and substituting the demand for reserves in the ample reserves region into the first-order condition yields the optimal level of reserves absent uncertainty, X^* :

$$X^* = \bar{X} - \frac{1}{\alpha^2 + \lambda} \left(\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}) \right) < \bar{X}.$$

The optimal level of reserves absent uncertainty increases less than one-for-one with the target level of reserves $\hat{X}$ and decreases with the target spread $\hat{S}$.⁹ The equilibrium spread implied by the optimal level of reserves absent uncertainty, S^* , is given by:

$$S^* = \frac{\alpha^2}{\alpha^2 + \lambda} \hat{S} + \frac{\lambda}{\alpha^2 + \lambda} S(\hat{X}),$$

where $S(\hat{X})$ is the point on the demand curve at $\hat{X}$.

The spread under the optimal supply of reserves absent uncertainty equals the target spread if $\lambda = 0$ and increases less than one-for-one with the target spread if $\lambda > 0$. Without uncertainty, the optimal spread lies between the target spread and the spread implied by the target level of reserves.

We now consider the other cases for the central bank targets. If the target for the

⁹The associated second-order condition for a minimum is $\alpha^2 + \lambda > 0$, which holds by assumption.

ex-ante supply of reserves is in the ample region ($\hat{X} < \bar{X}$), and the target spread is equal to the lower asymptote of the reserve demand curve ($\hat{S} = \bar{S}$; e.g., the Friedman rule absent frictions), the optimal supply of reserves and the optimal spread are the same as above. If the target level of reserves is in the abundant region ($\hat{X} > \bar{X}$), and the target spread is in the ample region ($\hat{S} > \bar{S}$), the optimal supply of reserves is the same as above if and only if the penalty for deviating from the target for reserves, λ , is sufficiently small; if λ is sufficiently large, the optimal supply of reserves is equal to the target level $\hat{X}$, and the optimal spread is $\bar{S}$.¹⁰ Obviously, if both targets are in the abundant region ($\hat{X} > \bar{X}$ and $\hat{S} = \bar{S}$), the central bank hits these targets exactly ($X^* = \hat{X}$ and $S^* = \hat{S}$).

In the following analyses, we focus on the more interesting cases where $\hat{X} < \bar{X}$, which can be interpreted as reflecting the concerns about an excessively large central bank balance sheet described in Section 2.4.

3.2 Expected Spread and Its Variance Under Uncertainty

We start by deriving the expected spread and its variance under uncertainty.¹¹ The derivations of all the results in this section are in Appendix A.2.

For a given level of reserves in the ample region, X , the cutoff value of the shock ϵ such that reserves become abundant, denoted by $\bar{\epsilon}(X)$, is given by:

$$\bar{\epsilon}(X) = X - \bar{X}.$$

For a given X , the probability of being in the abundant reserves region is then $\text{Prob}(\epsilon < \bar{\epsilon}(X)) = G(\bar{\epsilon}(X))$. To simplify notation, for the remainder of the paper, we use $\bar{\epsilon}$ to indicate

¹⁰The threshold is $\tilde{\lambda} = \alpha \frac{\hat{S} - \bar{S}}{\hat{X} - \bar{X}} \left(2 + \alpha \frac{\hat{X} - \bar{X}}{\hat{S} - \bar{S}} \right)^{-1}$.

¹¹The proofs underlying the results reported here build on methods developed for piecewise linear stochastic models developed in Mertens and Williams (2021) and Bok et al. (2025).

$\bar{\epsilon}(X)$.

The mean spread under uncertainty is

$$\mathbb{E}[S] = \bar{S} + \alpha \left(\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon} \right) \geq \max \left[\bar{S} - \alpha(X - \bar{X}), \bar{S} \right], \quad (4)$$

where $\mathcal{G}(x) = \int_{-\infty}^x G(u)du$ denotes the super-cumulative distribution function of ϵ , and the term on the right-hand side of the inequality is the spread in the deterministic case. The mean spread is greater than the deterministic one for all levels of reserves because there is an inherent asymmetry with regards to horizontal uncertainty: ϵ has no effect on the spread at high levels of reserves because its distribution is truncated along the horizontal portion of the reserve demand curve, but it does affect the spread in the sloped region of the curve. This asymmetry smooths the convex relationship between the mean spread and the level of reserves around the kink implied by the underlying piecewise-linear model and, as a result, the mean spread is higher for every reserve level. Figure 3 illustrates this result. In contrast, uncertainty about the vertical location of the demand curve (ν) has no effect on the mean spread, at any level of reserves. It is straightforward to show that the mean spread decreases with the ex-ante supply of reserves. In contrast to the deterministic case, however, the rate of decrease declines as reserves increase.

The variance of the spread is given by:

$$\text{Var}[S] = \sigma_\nu^2 - \alpha^2 \left(\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon} \right)^2 + \alpha^2 \left(\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon}) \right) \in [\sigma_\nu^2, \sigma_\nu^2 + \alpha^2 \sigma_\epsilon^2], \quad (5)$$

where $\mathbb{G}(x) = \int_{-\infty}^x \mathcal{G}(u)du$ denotes the super-super cumulative distribution function of ϵ . The effects of uncertainty about the horizontal location of the curve on the variance of the spread depend on the supply of reserves, as shown by the dependence on the cutoff $\bar{\epsilon}$. The effect of uncertainty about the vertical location, in contrast, is constant because vertical shifts move the spread by the same amount at every level of reserves. In the case without

horizontal shocks, the variance arises purely from vertical shocks, and $\text{Var}[S] = \sigma_v^2$.

Importantly, the variance of the spread decreases with the ex-ante supply of reserves. This result again arises from the floor of the demand curve: as the central bank supplies more reserves, the spread is more likely to be in the flat part of the demand curve, where it is insensitive to horizontal shocks. In particular, as reserves go to infinity, the variance of the spread converges to the variance of vertical shifts σ_v^2 . In contrast, as reserves go to negative infinity, the variance converges to $\sigma_v^2 + \alpha^2 \sigma_\epsilon^2$, which is the variance of the spread in the absence of a floor in the reserve demand curve. In practice, this limit corresponds to the case of a central bank operating far from the floor and deep into the sloped region of the reserve demand curve—such as in the monetary policy frameworks implemented by the central banks of most advanced economies before the GFC.

These results show that the central bank can influence the variability of the interest rate spread, even without a lending facility, through its provision of reserves. As we discuss in the next section, these features of the mean and variance of the spread have important implications for the optimal supply of reserves.

3.3 Optimal Supply of Reserves under Uncertainty

We now turn to the derivation of the optimal supply of reserves under uncertainty, in the absence of a lending facility. After substituting the reserve demand curve (1) into the optimization problem, the central bank's objective function is given by:

$$\min_X \frac{1}{2} \left\{ \left((\bar{S} - \hat{S})^2 + \sigma_v^2 \right) G(\bar{\epsilon}) + \mathbb{E} \left[\left(\bar{S} + v - \alpha(X - \bar{X} - \epsilon) - \hat{S} \right)^2 \middle| \epsilon > \bar{\epsilon} \right] + \lambda(X - \hat{X})^2 \right\}.$$

Note that, as we explain in greater detail below, certainty equivalence does *not* hold due to the truncation of the effect of horizontal shocks, ϵ , on the spread S .

Unlike the case of no uncertainty, the first-order condition is an implicit function

of the optimal reserves supply X^{**} that may have multiple solutions, some of which represent local minima and others local maxima depending on the sign of the second-order condition. That said, since the loss function is continuous and monotonically increases as reserves go to plus or minus infinity, there exists a global minimum. In our analysis, we rely on the second-order condition holding at the global minimum, while recognizing that there may be other local minima.

The key results regarding the effects of uncertainty on the optimal supply of reserves are summarized in the following proposition.

Proposition 1 (Uncertainty Increases Optimal Supply of Reserves)

*With joint uncertainty about ϵ and v , the optimal level of reserves, X^{**} , is characterized as follows*

$$X^{**} = X^* + \frac{\alpha^2}{\alpha^2 + \lambda} \mathcal{G}(\bar{\epsilon}) + \frac{\alpha}{\alpha^2 + \lambda} (\hat{S} - \bar{S}) G(\bar{\epsilon}) \geq X^*, \quad (6)$$

*which increases with $\hat{X}$ and decreases with $\hat{S}$. The associated mean spread under uncertainty, $\mathbb{E}S^{**}$, is given by:*

$$\mathbb{E}S^{**} = S^* + \alpha(\mathcal{G}(\bar{\epsilon}) + X^* - X^{**}) \leq S^*.$$

Proof: See Appendix A.3 for details on the derivations. □

Uncertainty about horizontal shifts in the reserve demand curve creates a precautionary motive to supply additional reserves. Uncertainty about horizontal shifts increases the optimal supply of reserves because the kink in the demand curve truncates the effect of reserves on the spread along the horizontal part of the curve. As a result, horizontal shifts have an asymmetric effect on the spread: on average, shifts to the right (i.e., higher demand) increase the spread more than shifts to the left (i.e., lower demand) decrease it. A larger supply of reserves has two effects: it reduces the deviation from the target spread caused by the increase of the mean spread due to uncertainty (see equation (4)), and it

reduces the overall variability of the spread (see equation (5)).

Uncertainty about horizontal shifts may increase or decrease the optimal mean spread due to two opposing effects: the direct effect of higher uncertainty, which increases the mean spread for any given value of reserves, and the indirect effect of a higher optimal level of reserves, which decreases the spread at the optimum. This result is important because it implies that it is not possible to judge the optimality of a central bank's reserve supply in response to changing market conditions based on the level of the spread.

Figure 3 illustrates the optimal choice of reserves and the corresponding mean policy rate spread under different uncertainty configurations. The black square denotes the optimal choices absent uncertainty. The red squares show the optimal choices when there is uncertainty about the kink (ϵ shocks).

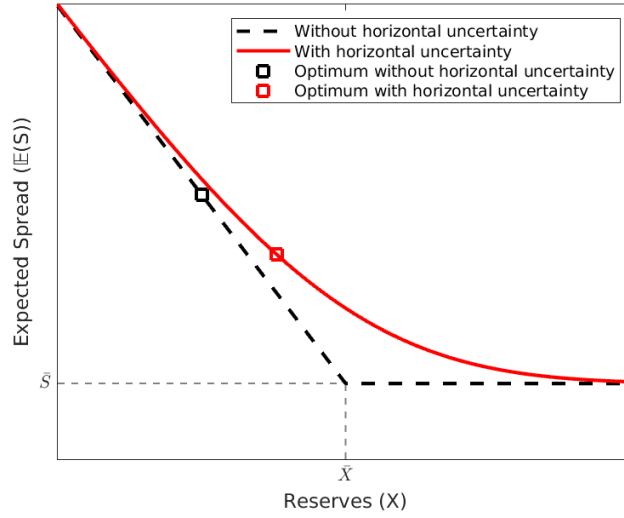
Finally, note that uncertainty about the floor of the demand curve (ν shocks) does not affect the mean spread or the optimal reserve supply. The reason is that vertical shocks simply add a constant term to the variance of the spread without affecting its mean (see equations (4) and (5)) and do not change how the supply of reserves affects the spread.

A direct implication of equation (6) is that an equally-sized joint increase in $\bar{X}$ and $\hat{X}$ yields a one-for-one increase in the optimal supply of reserves. Similarly, an equally-sized joint increase in $\bar{S}$ and $\hat{S}$ has no effect on the optimal supply of reserves. The reason is that, for rate control, only the difference between the target spread and the spread's lower asymptote matters.¹² As in the deterministic case, the optimal level of reserves is increasing in the target level of reserves (if $\lambda > 0$) and decreasing in the target spread.

Although we assume that the central bank targets a level of reserves in the ample reserves region ($\hat{X} < \bar{X}$), under certain conditions it can be optimal to set the level of reserves in the abundant region. Absent uncertainty about horizontal shifts in the reserve demand curve, if the central bank targets an ample reserve level, it is never optimal

¹²From equation (6), it is straightforward to show that $\frac{dX^{**}}{d\hat{S}} = -\frac{dX^{**}}{d\bar{S}}$; see Appendix A.3.

Figure 3: Expected Interest Rate Spread and Optimal Reserves. This figure shows the expected spread between the market-implied policy rate and the interest rate on reserve balances, both absent uncertainty and in the presence of different types of uncertainty about the reserve demand curve. The dashed black line represents the deterministic case without uncertainty about the location of the kink between ample and abundant reserves (ϵ shocks). The red line represents the case with uncertainty about the location of the kink. The squares represent the expected spreads at the optimal reserve levels in the different specifications: black for the deterministic case and red for the case with uncertainty about the location of the kink. In these figures, the zero-mean independent shocks ϵ and ν are assumed to be normally distributed. Note that the expected spread and optimal reserves when there is only uncertainty about the floor of the demand curve (ν shocks) are the same as in the deterministic case.



to supply abundant reserves. However, in the presence of this type of uncertainty, it is optimal to supply an abundant amount of reserves if the degree of uncertainty is sufficiently large; in particular if and only if the following condition is met:

$$-\alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) + (\alpha^2 + \sigma_\eta^2)\mathcal{G}(\bar{\epsilon}) - \lambda(\bar{X} - \hat{X}) > 0.$$

See Appendix A.3 for a derivation. Importantly, whether providing abundant reserves is optimal depends on the weight that the central bank puts on deviations from the target level of reserves, λ . A higher value of λ raises the hurdle to meet the condition for abundant reserves to be optimal.

3.4 Extensions

The basic model can be extended in a number of directions without loss of analytical tractability. We briefly consider two such extensions here and leave the details to the appendix. The first incorporates uncertainty about the slope of the reserve demand curve. The second adds a steeper part of the demand curve for low, or scarce, levels of reserves.

Uncertainty about the slope of the demand curve provides an additional reason to increase the supply of central bank reserves. It is optimal to guard against shocks to the slope by supplying an additional buffer of reserves, analogous to the literature on conservatism in the presence of multiplicative uncertainty started by Brainard (1967). Appendix A.4.1 shows this extension.

The addition of a region with a steeper slope also reinforces the precautionary motive to supply more reserves because shocks that push the level of reserves into the steeply-sloped part of the curve generate higher rate variability. As a result, the optimal supply of reserves is larger than in the otherwise equivalent two-region model under uncertainty. Moreover, the condition under which the central bank finds it optimal to supply abundant reserves is relaxed. Appendix A.4.2 shows this extension.

4 Optimal Policy With a Lending Facility

In this section, we show how the presence of fixed-rate lending facilities can alleviate the effect of uncertainty on the optimal supply of reserves by reducing the central bank's precautionary motive to supply more reserves. A lending facility prevents the interest rate spread from rising beyond a certain level, introducing a ceiling in the distribution of realized spreads. We derive both the conditions under which the optimal reserve supply is higher or lower than in implementation frameworks without a lending facility and the

optimal spread between the lending facility rate and the interest rate on reserves.¹³ As we explain below, the tradeoff between rate control and market disintermediation caused by facility usage makes the central bank problem non-trivial.

4.1 Expected Spread and Variance of the Spread

In the presence of a lending facility, the expected spread is given by:

$$\mathbb{E}[S] = \bar{S}_L - \alpha [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] \in [\bar{S}, \bar{S}_L], \quad (7)$$

where $\bar{\epsilon}_L = \bar{\epsilon} + \frac{\bar{S}_L - \bar{S}}{\alpha}$. The derivation of equation (7) is given in Appendix B.1. It is straightforward to show that the expected spread increases with the lending facility spread ($\bar{S}_L$) and decreases with the ex-ante supply of reserves. It converges to the lending facility spread from below as reserves go to minus infinity and to the lower asymptote of the reserve demand curve ($\bar{S}$) from above as reserves go to infinity. Moreover, there exists only one critical level of the ex-ante supply of reserves at which the expected spread is equal to the deterministic spread, with the expected spread being below the deterministic one for lower reserve levels and above it for higher reserve levels. If the distribution of horizontal uncertainty $g(\epsilon)$ is symmetric around zero, the intersection between the expected and deterministic spreads occurs at the midpoint between the two kinks in the reserve demand curve ($\bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha}$). The proofs of these results are in Appendix B.1.

The variance of the spread in the presence of a lending facility is given by:

$$\begin{aligned} \text{Var}[S] = & \sigma_v^2 + (\bar{S}_L - \bar{S})^2 - 2\alpha (\bar{S}_L - \bar{S}) \mathcal{G}(\bar{\epsilon}_L) - \alpha^2 \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\}^2 \\ & + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})]. \end{aligned} \quad (8)$$

¹³For the purpose of our analysis, the institutional details of the facility set-up, such as collateral eligibility and loan term, are not material.

See Appendix B.1 for a derivation. The variance in equation (8) increases with the lending facility spread and converges to the variance of vertical uncertainty σ_v^2 from above as the ex-ante supply of reserves go to minus or plus infinity.¹⁴ Moreover, if horizontal uncertainty is symmetric around zero, the variance is symmetric around the midpoint between the kinks in the reserve demand curve. Additionally, if the probability density function of horizontal uncertainty is also log-concave, the variance has a local maximum at the midpoint between the kinks. Although we cannot prove it formally, in all our simulations with symmetric single-peaked distributions of horizontal uncertainty, this local maximum is also the unique maximum.¹⁵ In other words, if the distribution of horizontal uncertainty is sufficiently well-behaved, the variance of the spread monotonically increases as the ex-ante supply of reserves goes from minus infinity to the midpoint between the kinks in the reserve demand curve and monotonically decreases above that point.¹⁶ The proofs of these results are in Appendix B.1.

4.2 Optimal Policy with a Lending Facility

To fix ideas, we start by analyzing optimal reserves provision with an exogenous lending facility rate. This case may be relevant if there are constraints on the central bank that prevent an optimal adjustment of the lending rate. In that case, the single choice facing the central bank is the amount of ex-ante reserves provided to the financial sector. We return to the joint decision of the lending rate and reserves below.

¹⁴Obviously, if the lending facility rate is equal to the interest rate on reserves, the variance is equal to σ_v^2 for all reserve levels. This is because, in this case, the reserve demand curve is a horizontal line at $\bar{S}_L = \bar{S}$.

¹⁵In our simulations, we use symmetric single-peaked density functions such as the Gaussian and Logistic distributions, which are log-concave and decay exponentially in the tails, and the Student's t -distribution, which is not log-concave and decays as a power law.

¹⁶If the distribution of horizontal uncertainty $g(\epsilon)$ is single-peaked but not symmetric, the maximum of the variance is between the kinks but not necessarily at the midpoint.

The first-order condition for the optimal level of reserves is given by:

$$\begin{aligned}
& (\alpha^2 + \lambda)(X - \bar{X}) + \lambda(\bar{X} - \hat{X}) + \alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) - \alpha^2 \mathcal{G}(\bar{\epsilon}) \\
& = -\alpha(1 - G(\bar{\epsilon}_L))(\bar{S}_L - \hat{S}) - \alpha^2(\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L) + \mu(1 - G(\bar{\epsilon}_L)) = 0,
\end{aligned} \tag{9}$$

where the left-hand side of this equation coincides with the first-order condition absent a lending facility. See Appendix B.2 for a derivation.

The right-hand side consists of two negative terms and a positive one. The negative terms, that call for lower ex-ante reserves provision, arise from the fact that the lending facility guards against large increases in the market spread. Since optimal reserves supply was higher under uncertainty because spreads could rise arbitrarily but cannot fall below the reserve demand floor, this channel is now mitigated through the lending facility. The positive term arises from the penalty placed on lending facility usage: a higher ex-ante reserves supply lowers the probability that the lending facility will be used. Taken together, equation (9) yields the following result.

Proposition 2 (Optimal Reserves Supply with Exogenous Lending Facility Rate)

For a given exogenous lending facility spread, under uncertainty about horizontal shifts in the demand for reserves, the optimal supply of reserves may be higher or lower than absent the facility. If the penalty associated with lending facility usage is sufficiently small, ex-ante optimal reserves supply is lower with an exogenous lending facility.

The existence of the lending facility reduces the precautionary motive to supply reserves to control rates because the facility introduces a cap on the market spread. However, if $\mu > 0$, the facility also introduces a precautionary motive to increase the supply of reserves to limit facility usage because the latter is costly for the central bank. If μ is sufficiently small, the first effect dominates, and the sum of the terms in the second line of first-order condition (9) is positive, implying that the optimal ex-ante supply of reserves is smaller

with a lending facility than without. If, however, μ is sufficiently large, the second effect dominates, and the optimal supply of reserves exceeds that absent a lending facility. The proofs of these results are in Appendix B.2.

We now analyze the optimal choice for the lending facility spread. The full optimization problem is described in equation (3), where the central bank jointly chooses the ex-ante supply of reserves, X , and the lending rate spread, $\bar{S}_L$. Both decision variables are chosen before the realization of the shock.

The resulting first-order condition for the optimal choice of X remains the same as in (9). In addition, the central bank also optimizes over the lending spread. Together, the first- and second-order optimality conditions for X and $\bar{S}_L$ yield the following result.

Proposition 3 (Optimal Choices with Endogenous Lending Facility Rate)

With joint uncertainty about ϵ and v , the central bank makes the following choices:

1. *The optimal choice of the lending rate spread, $\bar{S}_L^*$ is given by:*

$$\bar{S}_L^* = \hat{S} + \frac{\mu}{\alpha} \geq \hat{S},$$

which strictly increases with $\hat{S}$ and μ , and is independent of $\hat{X}$.

2. *The optimal ex-ante supply of reserves is characterized by the implicit function:*

$$(\alpha^2 + \lambda)(X - \bar{X}) + \lambda(\bar{X} - \hat{X}) + \alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) - \alpha^2 \mathcal{G}(\bar{\epsilon}) + \alpha^2(\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L) = 0;$$

as a result, the optimal ex ante supply of reserves is lower than the optimal supply absent a lending facility. Moreover, the optimal supply of reserves decreases with $\hat{S}$ and increases with $\hat{X}$ and with μ .

Proof: See Appendix B.2 for details of the derivations. □

The optimal lending spread is greater than or equal to the target spread, with equality in the limiting case of $\mu = 0$. If the central bank is indifferent to facility usage, it can maximize rate control by setting the facility spread equal to the target spread. This choice caps the market spread exactly at the target and eliminates upward deviations. If the central bank perceives a cost of facility usage, it is optimal to set a higher facility spread to reduce the likelihood that banks borrow from the facility. For the same reason, the optimal facility spread is increasing in the size of the penalty on facility usage: central banks facing a higher penalty optimally set a higher facility spread to induce lower borrowing levels.

When the central bank jointly sets the facility spread and the ex-ante supply of reserves, the optimal supply is lower than absent the facility. By choosing the lending rate spread optimally, the central bank minimizes the market-spread deviations from its target level with a lower reserve supply. Note that the optimal ex-ante supply of reserves increases with the penalty on facility usage, μ . There are two reasons for this result. First, if the penalty is higher, the central bank prefers to provide more reserves to reduce the likelihood of banks' borrowing from the facility. Second, a higher penalty increases the facility spread, which relaxes rate control; as a result, the central bank finds it optimal to offset this indirect effect by increasing the ex-ante supply of reserves.

5 Rate Control and Monetary Policy Frameworks

In this section, we compare the variability of the interbank market rate under different monetary policy implementation frameworks. We show that frameworks with scarce reserves and large spreads between the lending facility rate and the interest rate on reserve balances imply larger rate variability than frameworks with ample or abundant reserves. To illustrate this result, we use both the results of our model for the variance of the market spread under uncertainty and the experience of the United States over the last

two decades. We also discuss additional considerations that might influence the central bank optimal policy by affecting the weights λ and μ in the central bank objective function (3).

5.1 Rate Control: the U.S. Experience

Before the GFC, central bank reserves in the U.S. were kept scarce, and the spread between the lending facility rate and the interest rate on reserve balances—the so-called “corridor”—was large.¹⁷ In our model, this framework corresponds to a system where the target reserve level $\hat{X}$ is in the steep part of the reserve demand curve, and the target spread $\hat{S}$ is far from both the lower and the upper asymptotes of the curve. This type of framework can therefore be approximated by assuming that the demand for reserves is simply given by the negatively-sloped line in equation (2). Since the demand curve is linear, the central bank optimization problem in equation (3) is certainty equivalent, and the optimal supply of reserves coincides with the optimal supply absent uncertainty. In this limiting case, the variance of the optimal spread is¹⁸

$$\text{Var}[S] = \sigma_v^2 + \alpha^2 \sigma_\epsilon^2, \quad (10)$$

which exceeds not only the variance when a lending facility with a narrow corridor is present (equation (8)), but also the variance when the central bank operates a system of ample or abundant reserves, i.e., near or at the lower asymptote of the reserve demand curve, without a lending facility (equation (5)). The reason is that, when reserves are scarce, the distribution of interest rate spreads is wider and symmetric. As reserves increase, the demand curve flattens, truncating low realizations of the market spread.

¹⁷Reserves were not remunerated before the GFC. The Federal Reserve only started paying interest on reserve balances on October 9, 2008.

¹⁸This result can be derived by simply taking the limit of the variance of the spread without lending facility as reserves go to negative infinity, i.e., far from the curve’s lower asymptote.

Furthermore, as equation (10) shows, the variability of the market spread is independent of the level of reserves in traditional scarce systems with large corridors, in contrast to its variability in systems with narrow corridors or ample reserves (equations (5) and (8)).

Figure 4 shows the standard deviation of the spread between the federal funds rate and a measure of the target rate, as well as the level of reserves, before and after the GFC.¹⁹ The volatility of the spread averaged 0.10 percentage points in 2000–08, when reserves were scarce, and the corridor relatively large.²⁰ Consistent with the results from our model, as the supply of reserves increased and the Federal Reserve transitioned to a framework with ample reserves, volatility decreased substantially in 2009 and has fluctuated around 0.01 percentage points since then.

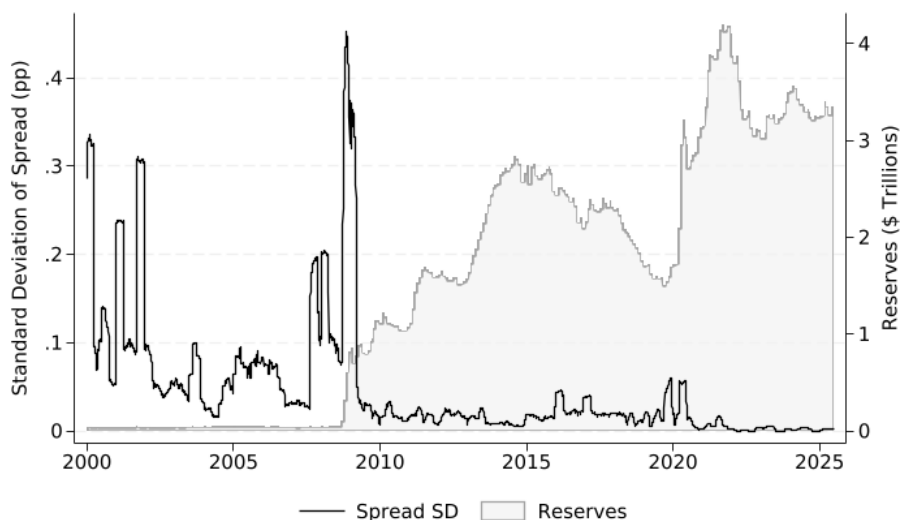
5.2 Other considerations

In addition to rate control (i.e., keeping the market spread close to its target level $\hat{S}$), other factors may influence the central bank choice of the optimal ex-ante supply of reserves. In our model, these factors are captured by the target reserve level $\hat{X}$, and their relative importance in the central bank optimization problem by the penalty weight assigned to deviations from this target, λ . Financial stability considerations are an important example. On the one hand, a level of reserves that satiates banks' demand for reserves supports the smooth functioning of financial markets by reducing the likelihood of a liquidity crunch in wholesale funding markets (Afonso et al., 2021; Copeland et al., 2025), improving the resilience of payment systems (Keister et al., 2008; Afonso et al., 2022; Goldstein et al., 2023), reducing the use of intraday overdrafts from the central bank (Kroeger et al., 2018),

¹⁹By construction, the variance of the market rate minus its target is equal to $\text{Var}(S - \hat{S}) = \text{Var}(S)$ in our model. Until December 2008, the Federal Open Market Committee set a target level for the federal funds rate; since then, it has set a target range. In our calculations, we use the mid-point of the range post 2008.

²⁰During 2000–08, central bank reserves (including vault cash) in the U.S. averaged \$58.6 billion (or 0.68% of bank assets), and the spread between the lending facility rate and the interest on reserve balances (which was zero) averaged 3.66 percentage points. From 2009 to June 2025, these averages were \$2.4 trillion (or 13.66% of bank assets) and 0.39 percentage points, respectively.

Figure 4: Bank Reserves and Variation in Daily Spread. The solid black line represents the standard deviation of the daily interest rate spread. The spread is the difference between the daily volume-weighted average federal funds rate and a measure of the target rate; consistent with the changes in monetary policy implementation during the sample period, the target rate is measured as the federal funds target rate for 01/01/2000–12/15/2008 and as the mid-point of the target range for 12/16/2008–06/30/2025. The 90-business-day windows (roughly a quarter) roll daily. The gray shaded area shows central bank reserves, including vault cash. Data are from January 2000 to June 2025. The daily volume-weighted average federal funds rate is publicly available from the Federal Reserve Economic Data, FRED (“EFFR,” 01/01/2000–02/29/2016) and from the Federal Reserve Bank of New York (03/01/2016–06/30/2025). The daily federal funds target rate, and the upper limit and lower limit of the federal funds target range are available from FRED (“DFEDTAR,” “DFEDTARU,” and “DFEDTARL”). Monthly data on reserves are available from FRED (“TOTRESNS”).



and discouraging excessive creation of runnable short-term debt by the private sector (Hanson et al., 2015; Greenwood et al., 2016).

On the other hand, large injections of reserves in the banking system can hinder financial stability if banks finance these balance-sheet expansions with liquidity commitments that are difficult to unwind during quantitative tightening, becoming more vulnerable to liquidity shocks and more dependent on central bank liquidity (Acharya and Rajan, 2024; Acharya et al., 2023). Moreover, if banks are subject to balance-sheet costs, large reserve

injections tighten these constraints and can therefore crowd out bank lending to the real economy (Martin et al., 2016; Diamond et al., 2020). Prolonged periods of high reserve levels can also cause political economy concerns due to large fluctuations in central bank net income and remittances to the government (Bindseil, 2016). Finally, a large ex ante supply of reserves discourages unsecured overnight money market activity, possibly hindering market discipline and price discovery and limiting the market's ability to redistribute reserves in times of stress (Borio, 2023).

Note that the same concern about market disintermediation also applies to standing lending facilities with a low interest rate spread relative to the interest rate on reserves: if the price of contingent central bank liquidity is attractive relative to the price of private liquidity, interbank market volume may decline. In our model, central banks more concerned about this issue would have a larger penalty weight on the usage of the lending facility, μ .

6 Conclusion

This paper analyzes the optimal supply of central bank reserves under uncertainty from the perspective of rate control and monetary policy implementation. Absent a lending facility, uncertainty about the level of reserves that satiates banks' demand for them increases the optimal supply of reserves. With a sufficiently high degree of uncertainty, it is optimal to supply a level of reserves that exceeds the satiation point even though the central bank targets the region below demand satiation; that is, the optimal supply of reserves lies in the abundant reserves region even if the central bank targets ample reserves.

With a lending facility, the optimal supply of reserves depends critically on the central bank preferences and constraints on the design of the facility. If the lending facility rate

spread can be chosen optimally and jointly with the ex-ante supply of reserves, the optimal ex ante supply is smaller than the one obtained absent a lending facility. In contrast, if there are frictions preventing the central bank from optimally setting the lending facility jointly with the ex ante supply of reserves, the optimal supply may be larger than absent the facility.

A key advantage of this analysis is that it is analytically tractable and can derive explicit expressions that convey the intuition. Potential extensions of the model include adding intertemporal aspects of the supply and demand for reserves, structural changes, and the role of experimentation and learning in the design of optimal policy. In addition, this model provides a flexible structure that can be empirically implemented, allowing for the estimation of the effects of uncertainty on the supply for reserves across different implementation frameworks.

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Appendix

A Proofs of Section 3 (without Lending Facility)

A.1 Optimal supply absent uncertainty

In this section, we derive the optimal supply of reserves absent uncertainty. We start by assuming that both central bank targets are in the ample region, i.e., $\hat{X} < \bar{X}$ and $\hat{S} > \bar{S}$. To derive the optimal supply in this case, we can simply take the first-order condition of the central bank's objective function in the ample reserves region:

$$\begin{aligned} 0 = \frac{\partial \mathcal{L}}{\partial X} &= \frac{1}{2} \{ -2\alpha [\bar{S} - \alpha(X^* - \bar{X}) - \hat{S}] + 2\lambda(X^* - \hat{X}) \} \\ &= (\alpha^2 + \lambda)X^* - \alpha(\bar{S} + \alpha\bar{X} - \hat{S}) - \lambda\hat{X} \\ &= (\alpha^2 + \lambda)(X^* - \hat{X}) + \alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}), \end{aligned}$$

which gives $X^* = \bar{X} - \frac{1}{\alpha^2 + \lambda}[\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X})] < \bar{X}$. The second-order condition is $\frac{\partial^2 \mathcal{L}}{\partial X^2} = \alpha^2 + \lambda > 0$, which holds by assumption.

Plugging the optimal supply of reserves absent uncertainty in the deterministic version of the reserve demand curve, we obtain the optimal spread absent uncertainty

$$S^* = \hat{S} - \frac{\lambda}{\alpha^2 + \lambda}(\hat{S} - \bar{S} - \alpha(\bar{X} - \hat{X})).$$

Since the spread at the level of reserves $\hat{X}$ is $S(\hat{X}) = \hat{S} - \alpha(X - \bar{X})$, we can rewrite S^* as

$$S^* = \hat{S} - \frac{\lambda}{\alpha^2 + \lambda}(\hat{S} - (\bar{S} - \alpha(\hat{X} - \bar{X}))) = \hat{S} - \frac{\lambda}{\alpha^2 + \lambda}(\hat{S} - S(\hat{X})) = \frac{\alpha^2}{\alpha^2 + \lambda}\hat{S} + \frac{\lambda}{\alpha^2 + \lambda}S(\hat{X}).$$

If $\hat{X} < \bar{X}$ and $\hat{S} = \bar{S}$, the optimal supply of reserves is the same as above. The reason is

that the central bank has no incentive to supply $X > \bar{X}$ to meet $\hat{S} = \bar{S}$ because the demand curve is flat at $\bar{S}$ for $X \geq \bar{X}$; the optimization problem can therefore be solved by taking the first-order condition in the ample region.

If $\hat{X} > \bar{X}$ and $\hat{S} = \bar{S}$, the problem is trivial: the central bank will supply exactly $\hat{X}$ because this choice sets both the squared deviation from the reserve target and the squared deviation from the spread target equal to zero.

If $\hat{X} > \bar{X}$ and $\hat{S} > \bar{S}$, the optimal solution is bang-bang, depending on the weight λ assigned by the central bank to deviations from the reserve target. If the solution is in the abundant region, it must be $\hat{X}$: any other level of abundant reserves would increase the squared deviations from the reserve target without reducing the squared deviation from the spread target (which is equal to $(\bar{S} - \hat{S})^2$ for all $X \geq \bar{X}$). If the solution is in the ample region, it must be equal to the solution of the first-order condition above. To identify the threshold λ that determines which solution is the global minimum, we just need to compare the values of the objective function at the two local minima:

$$\begin{aligned}\mathcal{L}(\hat{X}) &= \frac{1}{2} (\hat{S} - \bar{S})^2, \\ \mathcal{L}(X^*) &= \frac{1}{2} \left\{ \left[\bar{S} + \frac{\alpha}{\alpha^2 + \lambda} (\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X})) - \hat{S} \right]^2 + \lambda \left[\bar{X} - \frac{1}{\alpha^2 + \lambda} (\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X})) - \hat{X} \right]^2 \right\} \\ &= \frac{1}{2} (\hat{S} - \bar{S})^2 \frac{\lambda}{\alpha^2 + \lambda} \left(1 + \alpha \frac{\hat{X} - \bar{X}}{\hat{S} - \bar{S}} \right)^2.\end{aligned}$$

Supplying reserves exactly equal to $\hat{X}$ is optimal if and only if $\frac{\lambda}{\alpha^2 + \lambda} \left(1 + \alpha \frac{\hat{X} - \bar{X}}{\hat{S} - \bar{S}} \right)^2 > 1$. Since the left-hand side of this inequality is strictly increasing in λ , the inequality holds if and only if $\lambda > \alpha \frac{\hat{S} - \bar{S}}{\hat{X} - \bar{X}} \left(2 + \alpha \frac{\hat{X} - \bar{X}}{\hat{S} - \bar{S}} \right)^{-1}$. If λ is below this threshold, the optimal supply of reserves is in the ample region and is given by the first order condition above.

A.2 Mean spread and variance of the spread

In deriving the properties of the model under uncertainty, it is useful to define two functions related to the cumulative distribution function $G(\cdot)$. First, define the super-cumulative distribution function

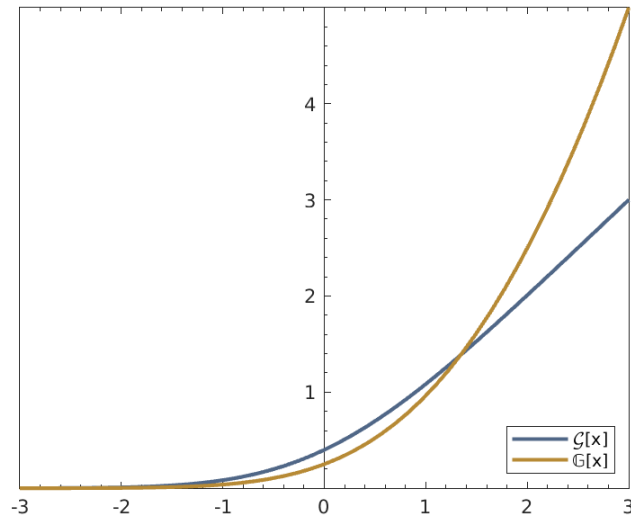
$$\mathcal{G}(a) = \int_{-\infty}^a G(\epsilon) d\epsilon \geq 0.$$

Since ϵ has zero mean, $\mathcal{G}(a) \geq a$ for all a . Second, define its integral

$$\mathbb{G}(a) = \int_{-\infty}^a \mathcal{G}(\epsilon) d\epsilon \geq 0,$$

which we refer to as the super-super cumulative distribution function. Note that $\mathbb{G}'(a) = \mathcal{G}(a)$ and $\mathcal{G}'(a) = G(a)$. Figure A.1 depicts the functions $\mathcal{G}$ and $\mathbb{G}$ for the standard normal distribution.

Figure A.1: **Super-cumulative Distribution Function of ϵ and its Integral.** This figure plots the super-cumulative distribution function $\mathcal{G}$ and its integral $\mathbb{G}$ for the standard normal distribution.



Based on this notation, the expected spread is

$$\begin{aligned}
\mathbb{E}S &= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S} + v) f(v) g(\epsilon) dv d\epsilon + \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} (\bar{S} + v - \alpha(X - \bar{X} - \epsilon)) f(v) g(\epsilon) dv d\epsilon \\
&= \bar{S} - \int_{\bar{\epsilon}}^{\infty} \alpha(X - \bar{X} - \epsilon) g(\epsilon) d\epsilon \\
&= \bar{S} - \alpha \int_{\bar{\epsilon}}^{\infty} (X - \bar{X} - \epsilon) g(\epsilon) d\epsilon = \bar{S} + \alpha(\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}) \geq \max[\bar{S} - \alpha(X - \bar{X}), \bar{S}],
\end{aligned}$$

where for the last inequality we used $\mathcal{G}(a) \geq 0$ and $\mathcal{G}(a) \geq a$ for all a . Its derivative is

$$\frac{d\mathbb{E}[S]}{dX} = -\alpha(1 - G(\bar{\epsilon})) \leq 0,$$

which shows that the mean spread decreases with reserves.

To derive the variance of the spread, we first derive the expected value of the squared spread as follows

$$\begin{aligned}
\mathbb{E}[S^2] &= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S} + v)^2 f(v) g(\epsilon) dv d\epsilon + \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} [\bar{S} + v - \alpha(X - \bar{X} - \epsilon)]^2 f(v) g(\epsilon) dv d\epsilon \\
&= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S}^2 + 2v\bar{S} + v^2) f(v) g(\epsilon) dv d\epsilon + \\
&\quad \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} [(\bar{S} + v)^2 - 2\alpha(\bar{S} + v)(\bar{\epsilon} - \epsilon) + \alpha^2(\bar{\epsilon} - \epsilon)^2] f(v) g(\epsilon) dv d\epsilon \\
&= \bar{S}^2 G(\bar{\epsilon}) + 2\bar{S} \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} v f(v) dv \right) g(\epsilon) d\epsilon + \sigma_v^2 G(\bar{\epsilon}) + \bar{S}^2 [1 - G(\bar{\epsilon})] + \\
&\quad 2\bar{S} \int_{\bar{\epsilon}}^{\infty} \left(\int_{-\infty}^{\infty} v f(v) dv \right) g(\epsilon) d\epsilon + \sigma_v^2 [1 - G(\bar{\epsilon})] - 2\alpha\bar{S} \int_{\bar{\epsilon}}^{\infty} \left(\int_{-\infty}^{\infty} f(v) dv \right) (\bar{\epsilon} - \epsilon) g(\epsilon) d\epsilon - \\
&\quad 2\alpha \int_{\bar{\epsilon}}^{\infty} \left(\int_{-\infty}^{\infty} v f(v) dv \right) (\bar{\epsilon} - \epsilon) g(\epsilon) d\epsilon + \alpha^2 \int_{\bar{\epsilon}}^{\infty} \left(\int_{-\infty}^{\infty} f(v) dv \right) (\bar{\epsilon} - \epsilon)^2 g(\epsilon) d\epsilon \\
&= \sigma_v^2 + \bar{S}^2 - 2\alpha\bar{S} \left[\bar{\epsilon}(1 - G(\bar{\epsilon})) - \int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon \right] + \\
&\quad \alpha^2 \left[\bar{\epsilon}^2 (1 - G(\bar{\epsilon})) - 2\bar{\epsilon} \int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon + \int_{\bar{\epsilon}}^{\infty} \epsilon^2 g(\epsilon) d\epsilon \right]
\end{aligned}$$

where the first equality holds because ϵ and ν are independent, and the fourth equality holds because ν has zero mean.

Since ϵ has zero mean and finite variance σ_ϵ^2 , using integration by parts, we can write

$$\begin{aligned}\int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon &= - \int_{-\infty}^{\bar{\epsilon}} \epsilon g(\epsilon) d\epsilon = -\bar{\epsilon} G(\bar{\epsilon}) + \int_{-\infty}^{\bar{\epsilon}} G(\epsilon) d\epsilon = -\bar{\epsilon} G(\bar{\epsilon}) + \mathcal{G}(\bar{\epsilon}), \\ \int_{\bar{\epsilon}}^{\infty} \epsilon^2 g(\epsilon) d\epsilon &= \sigma_\epsilon^2 - \int_{-\infty}^{\bar{\epsilon}} \epsilon^2 g(\epsilon) d\epsilon = \sigma_\epsilon^2 - \bar{\epsilon}^2 G(\bar{\epsilon}) + \int_{-\infty}^{\bar{\epsilon}} 2\epsilon G(\epsilon) d\epsilon \\ &= \sigma_\epsilon^2 - \bar{\epsilon}^2 G(\bar{\epsilon}) + 2 \left(\bar{\epsilon} \mathcal{G}(\bar{\epsilon}) - \int_{-\infty}^{\bar{\epsilon}} \mathcal{G}(\epsilon) d\epsilon \right) = \sigma_\epsilon^2 - \bar{\epsilon}^2 G(\bar{\epsilon}) + 2\bar{\epsilon} \mathcal{G}(\bar{\epsilon}) - 2\mathbb{G}(\bar{\epsilon}),\end{aligned}$$

where we used $\lim_{x \rightarrow -\infty} \mathcal{G}(x) = \lim_{x \rightarrow -\infty} \mathbb{G}(x) = 0$.

Using these expressions, we can write

$$\begin{aligned}\mathbb{E}[S^2] &= \sigma_\nu^2 + \bar{S}^2 - 2\alpha \bar{S} \left[\bar{\epsilon} (1 - G(\bar{\epsilon})) - \int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon \right] + \\ &\quad \alpha^2 \left[\bar{\epsilon}^2 (1 - G(\bar{\epsilon})) - 2\bar{\epsilon} \int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon + \int_{\bar{\epsilon}}^{\infty} \epsilon^2 g(\epsilon) d\epsilon \right] \\ &= \sigma_\nu^2 + \bar{S}^2 - 2\alpha \bar{S} [\bar{\epsilon} - \mathcal{G}(\bar{\epsilon})] + \alpha^2 \{ \bar{\epsilon}^2 [1 - G(\bar{\epsilon})] - 2\bar{\epsilon} [-\bar{\epsilon} G(\bar{\epsilon}) + \mathcal{G}(\bar{\epsilon})] + \sigma_\epsilon^2 - \bar{\epsilon}^2 G(\bar{\epsilon}) + \\ &\quad 2\bar{\epsilon} \mathcal{G}(\bar{\epsilon}) - 2\mathbb{G}(\bar{\epsilon}) \} \\ &= \sigma_\nu^2 + \bar{S}^2 + 2\alpha \bar{S} [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}] + \alpha^2 [\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon})].\end{aligned}$$

Putting together this expression with the one for the expected spread, we obtain the variance of the spread as

$$\begin{aligned}\text{Var}[S] &= \mathbb{E}[S^2] - (\mathbb{E}[S])^2 = \sigma_\nu^2 + \bar{S}^2 + 2\alpha \bar{S} [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}] + \alpha^2 [\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon})] - [\bar{S} + \alpha (\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon})]^2 \\ &= \sigma_\nu^2 - \alpha^2 [\mathcal{G}(\epsilon) - \epsilon]^2 + \alpha^2 [\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon})].\end{aligned}$$

Taking the derivative of this expression with respect to the level of reserves X , we get

$$\begin{aligned}\frac{d\text{Var}[S]}{dX} &= -2\alpha^2(\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}) + \alpha^2(2\bar{\epsilon} - 2\mathcal{G}(\bar{\epsilon})) \\ &= -2\alpha^2 G(\bar{\epsilon})(\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}) < 0,\end{aligned}$$

where the inequality follows from $\mathcal{G}(\bar{\epsilon}) > \bar{\epsilon}$ for all reserve values.

To prove the limiting behavior of $\text{Var}[S(X)]$ as X goes to plus and minus infinity, we use the following equalities, which are derived using integration by parts:

$$\begin{aligned}\mathcal{G}(x) &= \int_{-\infty}^x G(u)du = xG(x) - \int_{-\infty}^x u g(u)du, \\ \mathbb{G}(x) &= \int_{-\infty}^x \mathcal{G}(u)du = x\mathcal{G}(x) - \frac{x^2}{2}G(x) + \int_{-\infty}^x \frac{u^2}{2}g(u)du.\end{aligned}$$

Plugging these equalities in the expression for the variance above, we obtain:

$$\begin{aligned}\lim_{X \rightarrow \infty} \text{Var}[S] &= \sigma_v^2 + \lim_{X \rightarrow \infty} \left\{ -\alpha^2 [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]^2 + \alpha^2 [\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon})] \right\} \\ &= \sigma_v^2 + \alpha^2 \sigma_\epsilon^2 + \\ &\quad \lim_{X \rightarrow \infty} \left\{ -\alpha^2 \left[\bar{\epsilon}G(\bar{\epsilon}) - \int_{-\infty}^{\bar{\epsilon}} u g(u)du - \bar{\epsilon} \right]^2 + \alpha^2 \bar{\epsilon}^2 - \right. \\ &\quad \left. 2\alpha^2 \left[\bar{\epsilon} \left(\bar{\epsilon}G(\bar{\epsilon}) - \int_{-\infty}^{\bar{\epsilon}} u g(u)du \right) - \frac{\bar{\epsilon}^2}{2}G(\bar{\epsilon}) + \int_{-\infty}^{\bar{\epsilon}} \frac{u^2}{2}g(u)du \right] \right\} \\ &= \sigma_v^2 + \alpha^2 \sigma_\epsilon^2 + \lim_{X \rightarrow \infty} \left\{ \alpha^2 \bar{\epsilon}^2 - \alpha^2 \bar{\epsilon}^2 - \alpha^2 \sigma_\epsilon^2 \right\} = \sigma_v^2,\end{aligned}$$

where we used $\lim_{x \rightarrow \infty} G(x) = 1$ and the fact that ϵ has zero mean and finite variance.

Similarly,

$$\begin{aligned}\lim_{X \rightarrow -\infty} \text{Var}[S] &= \sigma_v^2 + \lim_{X \rightarrow \infty} \left\{ -\alpha^2 [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]^2 + \alpha^2 [\bar{\epsilon}^2 + \sigma_\epsilon^2 - 2\mathbb{G}(\bar{\epsilon})] \right\} \\ &= \sigma_v^2 + \alpha^2 \sigma_\epsilon^2 + \lim_{X \rightarrow \infty} \left\{ -\alpha^2 \bar{\epsilon}^2 + \alpha^2 \bar{\epsilon}^2 \right\} = \sigma_v^2 + \alpha^2 \sigma_\epsilon^2,\end{aligned}$$

where we used the fact that $\lim_{x \rightarrow -\infty} \mathcal{G}(x) = \lim_{x \rightarrow -\infty} \mathbb{G}(x) = 0$.

A.3 Optimal supply of reserves under uncertainty

The first-order condition of the central bank minimization problem (3) is

$$\begin{aligned}& \int_{-\infty}^{\infty} (\bar{S} + v - \hat{S})^2 f(v) dv g(\bar{\epsilon}) \frac{d\bar{\epsilon}}{dX} \\ & - \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\bar{S} + v - \alpha(X^{**} - \bar{X} - \bar{\epsilon}) - \hat{S} \right)^2 f(v) dv g(\bar{\epsilon}) \frac{d\bar{\epsilon}}{dX} \\ & + \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} -\alpha(\bar{S} + v - \alpha(X^{**} - \bar{X} - \epsilon) - \hat{S}) f(v) g(\epsilon) dv d\epsilon + \lambda(X^{**} - \hat{X}) = 0,\end{aligned}$$

where the first two terms cancel out when substituting in $\bar{\epsilon} = X^{**} - \bar{X}$. As a result, the first-order condition can be simplified to

$$\int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} -\alpha(\bar{S} + v - \alpha(X^{**} - \bar{X} - \epsilon) - \hat{S}) f(v) g(\epsilon) dv d\epsilon + \lambda(X^{**} - \hat{X}) = 0. \quad (\text{A.1})$$

Using the representation of $\int_{\bar{\epsilon}}^{\infty} \epsilon g(\epsilon) d\epsilon$ introduced in Appendix A.2 and the fact that v has zero mean, we can simplify the first-order condition in equation (A.1) to

$$\begin{aligned}0 &= \int_{\bar{\epsilon}}^{\infty} -\alpha(\bar{S} - \alpha(X^{**} - \bar{X} - \epsilon) - \hat{S}) g(\epsilon) d\epsilon + \lambda(X^{**} - \hat{X}) \\ &= (1 - G(\bar{\epsilon})) \left(-\alpha(\bar{S} - \hat{S}) + \alpha^2(X^{**} - \bar{X}) \right) - \int_{\bar{\epsilon}}^{\infty} \alpha^2 \epsilon g(\epsilon) d\epsilon + \lambda(X^{**} - \hat{X}) \\ &= (1 - G(\bar{\epsilon})) \left(-\alpha(\bar{S} - \hat{S}) + \alpha^2(X^{**} - \bar{X}) \right) + \alpha^2(\bar{\epsilon}G(\bar{\epsilon}) - \mathcal{G}(\bar{\epsilon})) + \lambda(X^{**} - \hat{X}).\end{aligned}$$

Using the definition $\bar{\epsilon} = X^{**} - \bar{X}$, we get

$$-\alpha(1 - G(\bar{\epsilon}))(\bar{S} - \hat{S}) + (\alpha^2 + \lambda)(X^{**} - \bar{X}) - \alpha^2 \mathcal{G}(\bar{\epsilon}) + \lambda(\bar{X} - \hat{X}) = 0,$$

which we can rearrange as

$$X^{**} = \bar{X} - \frac{\alpha}{\alpha^2 + \lambda}(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) - \frac{\lambda}{\alpha^2 + \lambda}(\bar{X} - \hat{X}) + \frac{\alpha^2}{\alpha^2 + \lambda}\mathcal{G}(\bar{\epsilon}),$$

or alternatively as

$$X^{**} = X^* + \frac{\alpha}{\alpha^2 + \sigma_\eta^2 + \lambda}G(\bar{\epsilon})(\hat{S} - \bar{S}) + \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda}\mathcal{G}(\bar{\epsilon}). \quad (\text{A.2})$$

Finally, differentiating the left-hand side of the first-order condition in equation (A.1) yields the following second-order condition

$$\begin{aligned} \Gamma &\equiv \frac{d}{dX} \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} -\alpha(\bar{S} + v - \alpha(X - \bar{X} - \epsilon) - \hat{S})f(v)g(\epsilon)dv d\epsilon + \lambda(X - \hat{X}) \\ &= \int_{-\infty}^{\infty} \alpha(\bar{S} + v - \hat{S})f(v)g(\bar{\epsilon})dv + \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} (\alpha + \eta)^2 f(v)g(\epsilon)dv d\epsilon + \lambda \\ &= -\alpha(\hat{S} - \bar{S})g(\bar{\epsilon}) + (1 - G(\bar{\epsilon}))\alpha^2 + \lambda > 0. \end{aligned} \quad (\text{A.3})$$

To derive the comparative statics of the optimal supply of reserves under uncertainty X^{**} with respect to the target level of reserves and the target spread, we apply the implicit function theorem. Let $F(X^{**}) = 0$ be the first-order condition derived above; it is straightforward to show that

$$\begin{aligned} \frac{dX^{**}}{d\hat{X}} &= - \left(\frac{\partial F / \partial \hat{X}}{\partial F / \partial X} \right) \Big|_{X=X^{**}}^{-1} = \frac{\lambda}{\Gamma} \geq 0, \text{ and} \\ \frac{dX^{**}}{d\hat{S}} &= - \left(\frac{\partial F / \partial \hat{S}}{\partial F / \partial X} \right) \Big|_{X=X^{**}}^{-1} = - \frac{\alpha(1 - G(\bar{\epsilon}))}{\Gamma} \leq 0, \end{aligned}$$

where $\Gamma > 0$ is the second-order condition.

Using the formula for the expected spread under uncertainty derived in Appendix A.2, we can write the expected spread at the optimal reserve level X^{**} as

$$\begin{aligned}\mathbb{E}S^{**} &= \bar{S} - \alpha(X^{**} - \bar{X}) + \alpha\mathcal{G}(\bar{\epsilon}) \\ &= \bar{S} - \alpha(X^* + \Delta X^{**} - \bar{X}) + \alpha\mathcal{G}(\bar{\epsilon}) \\ &= S^* + \alpha(\mathcal{G}(\bar{\epsilon}) - \Delta X^{**}),\end{aligned}$$

where $\Delta X^{**} = X^{**} - X^* \geq 0$. Since $X^* \leq \bar{X}$, we have $\Delta X^{**} \geq X^{**} - \bar{X} = \bar{\epsilon}$. Since $\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon} \geq 0$, we cannot sign $\mathcal{G}(\bar{\epsilon}) - \Delta X^{**}$. Thus, we get $\mathbb{E}S^{**} \leq S^*$.

Substituting in for X^{**} delivers

$$\begin{aligned}\mathbb{E}S^{**} &= \bar{S} - \frac{\alpha}{\alpha^2 + \lambda} \left(-\alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) - \lambda(\bar{X} - \hat{X}) + \alpha^2\mathcal{G}(\bar{\epsilon}) \right) + \alpha\mathcal{G}(\bar{\epsilon}) \\ &= \bar{S} - \frac{\alpha^2}{\alpha^2 + \lambda}(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) + \frac{\alpha\lambda}{\alpha^2 + \lambda}(\bar{X} - \hat{X}) - \frac{\alpha^3}{\alpha^2 + \lambda}\mathcal{G}(\bar{\epsilon}) + \alpha\mathcal{G}(\bar{\epsilon}) \\ &= \bar{S} - \frac{\alpha^2}{\alpha^2 + \lambda}(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) + \frac{\alpha\lambda}{\alpha^2 + \lambda}(\bar{X} - \hat{X}) + \frac{\alpha\lambda}{\alpha^2 + \lambda}\mathcal{G}(\bar{\epsilon}).\end{aligned}$$

A.4 Extensions

A.4.1 Uncertainty about the slope

With uncertainty about the slope of the demand curve, the reserve demand curve takes the form

$$S = \begin{cases} \bar{S} + \nu - (\alpha + \eta)(X - \bar{X} - \epsilon) & \text{if } X < \bar{X} + \epsilon, \\ \bar{S} + \nu & \text{else,} \end{cases}$$

where η is a shock to the slope that is independent from the horizontal and vertical shocks.

The analogous result to Proposition 1 is the following proposition.

Proposition 4 (Uncertainty about the Slope Increases the Optimal Supply of Reserves)

In the case of joint uncertainty about ϵ , ν , and η , the optimal level of reserves, X^{**} , is given by:

$$\begin{aligned} X^{**} = X^* &+ \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda} \mathcal{G}(\bar{\epsilon}) + \frac{\alpha}{\alpha^2 + \sigma_\eta^2 + \lambda} (\hat{S} - \bar{S}) G(\bar{\epsilon}) \\ &+ \frac{\sigma_\eta^2}{(\alpha^2 + \lambda)(\alpha^2 + \sigma_\eta^2 + \lambda)} \left(\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}) \right) \geq X^*, \end{aligned}$$

which increases with $\hat{X}$ and with σ_η^2 and decreases with $\hat{S}$.

The remainder of this section shows the proof of this proposition. The first-order condition of the problem with uncertainty about the slope is

$$\begin{aligned} &\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\bar{S} + \nu - \hat{S})^2 f(\nu) h(\eta) d\nu d\eta g(\bar{\epsilon}) \frac{d\bar{\epsilon}}{dX} \\ &- \frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\bar{S} + \nu - (\alpha + \eta)(X^{**} - \bar{X} - \bar{\epsilon}) - \hat{S} \right)^2 f(\nu) h(\eta) d\nu d\eta g(\bar{\epsilon}) \frac{d\bar{\epsilon}}{dX} \\ &+ \int_{-\infty}^{\infty} \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} -(\alpha + \eta)(\bar{S} + \nu - (\alpha + \eta)(X^{**} - \bar{X} - \epsilon) - \hat{S}) f(\nu) g(\epsilon) h(\eta) d\nu d\epsilon d\eta + \lambda(X^{**} - \hat{X}) = 0, \end{aligned}$$

where the first two terms cancel out when substituting in $\bar{\epsilon} = X^{**} - \bar{X}$. As a result, the first-order condition can be simplified to

$$\int_{-\infty}^{\infty} \int_{\bar{\epsilon}}^{\infty} \int_{-\infty}^{\infty} -(\alpha + \eta)(\bar{S} + \nu - (\alpha + \eta)(X^{**} - \bar{X} - \epsilon) - \hat{S}) f(\nu) g(\epsilon) h(\eta) d\nu d\epsilon d\eta + \lambda(X^{**} - \hat{X}) = 0.$$

Using the same steps as in Appendix A.3, we simplify the first-order condition to

$$\begin{aligned} 0 &= \int_{-\infty}^{\infty} \int_{\bar{\epsilon}}^{\infty} -(\alpha + \eta)(\bar{S} - (\alpha + \eta)(X^{**} - \bar{X} - \epsilon) - \hat{S}) g(\epsilon) h(\eta) d\epsilon d\eta + \lambda(X^{**} - \hat{X}) \\ &= (1 - G(\bar{\epsilon})) \left(-\alpha(\bar{S} - \hat{S}) + (\alpha^2 + \sigma_\eta^2)(X^{**} - \bar{X}) \right) - \int_{-\infty}^{\infty} \int_{\bar{\epsilon}}^{\infty} (\alpha + \eta)^2 \epsilon g(\epsilon) h(\eta) d\epsilon d\eta + \lambda(X^{**} - \hat{X}) \\ &= (1 - G(\bar{\epsilon})) \left(-\alpha(\bar{S} - \hat{S}) + (\alpha^2 + \sigma_\eta^2)(X^{**} - \bar{X}) \right) + (\alpha^2 + \sigma_\eta^2)(\bar{\epsilon} G(\bar{\epsilon}) - \mathcal{G}(\bar{\epsilon})) + \lambda(X^{**} - \hat{X}), \end{aligned}$$

where we used the fact that ϵ , ν , and η have zero mean. Using the definition $\bar{\epsilon} = X^{**} - \bar{X}$, we get

$$-\alpha(1 - G(\bar{\epsilon}))(\bar{S} - \hat{S}) + (\alpha^2 + \sigma_\eta^2 + \lambda)(X^{**} - \bar{X}) - (\alpha^2 + \sigma_\eta^2)\mathcal{G}(\bar{\epsilon}) + \lambda(\bar{X} - \hat{X}) = 0,$$

which we can rearrange as

$$X^{**} = \bar{X} - \frac{\alpha}{\alpha^2 + \sigma_\eta^2 + \lambda}(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) - \frac{\lambda}{\alpha^2 + \sigma_\eta^2 + \lambda}(\bar{X} - \hat{X}) + \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda}\mathcal{G}(\bar{\epsilon}).$$

From this equation, using the implicit function theorem, we obtain $\frac{dX^{**}}{d\hat{X}} = \frac{\lambda}{\Gamma(\alpha^2 + \sigma_\eta^2 + \lambda)} \geq 0$,

$$0, \frac{dX^{**}}{d\hat{S}} = -\frac{\alpha(1 - G(\bar{\epsilon}))}{\Gamma(\alpha^2 + \sigma_\eta^2 + \lambda)} \leq 0, \text{ and } \frac{dX^{**}}{d\sigma_\eta^2} = \frac{\alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}) + \lambda\mathcal{G}(\bar{\epsilon})}{\Gamma(\alpha^2 + \sigma_\eta^2 + \lambda)^2} \geq 0,$$

where $\Gamma > 0$ is the second-order condition for the optimal supply to be a minimum of the objective function.

Using $X^* = \bar{X} - \frac{1}{\alpha^2 + \lambda} \left[\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}) \right]$, we can write the optimal supply as

$$\begin{aligned} X^{**} &= X^* + \frac{\alpha(\hat{S} - \bar{S})}{\alpha^2 + \lambda} - \frac{\alpha(1 - G(\bar{\epsilon}))(\hat{S} - \bar{S})}{\alpha^2 + \sigma_\eta^2 + \lambda} + \frac{\lambda(\bar{X} - \hat{X})}{\alpha^2 + \lambda} - \frac{\lambda(\bar{X} - \hat{X})}{\alpha^2 + \sigma_\eta^2 + \lambda} + \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda}\mathcal{G}(\bar{\epsilon}) \\ &= X^* + \frac{\alpha(\hat{S} - \bar{S})}{(\alpha^2 + \lambda)(\alpha^2 + \sigma_\eta^2 + \lambda)} \left[\alpha^2 + \sigma_\eta^2 + \lambda - (\alpha^2 + \lambda)(1 - G(\bar{\epsilon})) \right] + \\ &\quad \frac{\lambda(\bar{X} - \hat{X})}{(\alpha^2 + \lambda)(\alpha^2 + \sigma_\eta^2 + \lambda)} \left[\alpha^2 + \sigma_\eta^2 + \lambda - \alpha^2 - \lambda \right] + \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda}\mathcal{G}(\bar{\epsilon}) \\ &= X^* + \frac{\alpha G(\bar{\epsilon})(\hat{S} - \bar{S})}{\alpha^2 + \sigma_\eta^2 + \lambda} + \frac{\sigma_\eta^2 \left[\alpha(\hat{S} - \bar{S}) + \lambda(\bar{X} - \hat{X}) \right]}{(\alpha^2 + \lambda)(\alpha^2 + \sigma_\eta^2 + \lambda)} + \frac{\alpha^2 + \sigma_\eta^2}{\alpha^2 + \sigma_\eta^2 + \lambda}\mathcal{G}(\bar{\epsilon}) \geq X^*. \end{aligned}$$

A.4.2 Including a region of scarce reserves

In this section, we derive the optimal supply of reserves in the presence of a third region, in which the slope of the reserve demand curve is steeper than in the ample reserves region; we refer to this region as the region of scarce reserves. Specifically, we introduce a second kink in the demand curve at $\bar{X}_2 < \bar{X}_1$, where $\bar{X}_1$ now denotes the cutoff between ample and abundant reserves (formerly $\bar{X}$), and $\bar{X}_2$ represents the cutoff between scarce and ample reserves.

The resulting reserve demand curve is given by:

$$S = \begin{cases} \bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \beta(X - \bar{X}_2 - \epsilon) & \text{if } \epsilon > X - \bar{X}_2, \\ \bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) & \text{if } X - \bar{X}_2 \geq \epsilon > X - \bar{X}_1, \\ \bar{S} + \nu & \text{if } \epsilon < X - \bar{X}_1, \end{cases}$$

where $\beta \geq 0$ is the increase in the slope in the scarce reserves region relative to that in ample reserves region (α). Absent any type of uncertainty, the spread is equal to $\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) > \bar{S}$ at the kink between scarce and ample reserves, $\bar{X}_2$.

For simplicity, we assume that the central bank targets the ample region for both the spread and reserves (i.e., $\bar{S} < \hat{S} < \bar{S} + \alpha(\bar{X}_1 - \bar{X}_2)$ and $\bar{X}_1 < \hat{X} < \bar{X}_2$), so that the optimal supply of reserves absent uncertainty lies in the ample reserves region. Specifically, in the absence of uncertainty, the equation for the optimal supply of reserves is identical to that in the two-region model (with $\bar{X}_1$ replacing $\bar{X}$). In the presence of uncertainty, the first-order condition for the optimal supply of reserves, X^{**} , yields the following proposition.

Proposition 5 (Higher Optimal Supply of Reserves with a Scarce Reserves Region)

The existence of the scarce reserves region increases the optimal supply of reserves under uncertainty:

$$X^{***} = X^{**} + \frac{\beta(1 - G(\bar{\epsilon}_2))}{\alpha^2 + \lambda} \left(\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) - \hat{S} \right) \geq X^{**},$$

where $\bar{\epsilon}_2 = \bar{\epsilon}(\bar{X}_2)$.

To prove the results above, we proceed as follows. When the demand curve includes a region of scarce reserves, the central bank's optimization problem is given by

$$\begin{aligned} \min_X & \int_{-\infty}^{\bar{\epsilon}_1} \int_{-\infty}^{\infty} (\bar{S} + \nu - \hat{S})^2 f(\nu) g(\epsilon) d\nu d\epsilon + \int_{\bar{\epsilon}_1}^{\bar{\epsilon}_2} \int_{-\infty}^{\infty} (\bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \hat{S})^2 f(\nu) g(\epsilon) d\nu d\epsilon + \\ & + \int_{\bar{\epsilon}_2}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \beta(X - \bar{X}_2 - \epsilon) - \hat{S} \right)^2 f(\nu) g(\epsilon) d\nu d\epsilon \\ & + \lambda(X - \hat{X})^2. \end{aligned}$$

Denote the objective function in the two-region model by $\mathcal{L}_1$, the state vector by $\theta = \{\epsilon, \nu\}$, the state space by Θ , and the joint density function by $\gamma(\theta)$. We further denote the partition of the state space for ϵ between $\bar{\epsilon}_2$ and ∞ (with all other variables unrestricted) by Θ_2 . Then, we use that

$$\begin{aligned} & \int_{\bar{\epsilon}_2}^{\infty} \int_{-\infty}^{\infty} \left(\bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \beta(X - \bar{X}_2 - \epsilon) - \hat{S} \right)^2 f(\nu) g(\epsilon) d\nu d\epsilon \\ & = \int_{\bar{\epsilon}_2}^{\infty} \int_{-\infty}^{\infty} (\bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \hat{S})^2 f(\nu) g(\epsilon) d\nu d\epsilon + \\ & + \int_{\theta \in \Theta_2} \left(-2(\bar{S} + \nu - \alpha(X - \bar{X}_1 - \epsilon) - \hat{S})\beta(X - \bar{X}_2 - \epsilon) + \beta^2(X - \bar{X}_2 - \epsilon)^2 \right) \gamma(\theta) d\theta. \end{aligned}$$

Rearranging the terms, we obtain the objective function

$$\begin{aligned}\mathcal{L}_1 &+ \int_{\theta \in \Theta_2} \left(-2(\bar{S} + \nu - \hat{S})\beta(X - \bar{X}_2 - \epsilon) + 2\alpha\beta(X - \bar{X}_1 - \epsilon)(X - \bar{X}_2 - \epsilon) + \beta^2(X - \bar{X}_2 - \epsilon)^2 \right) \gamma(\theta) d\theta \\ &= \mathcal{L}_1 + \int_{\theta \in \Theta_2} \left(2(\hat{S} - \bar{S})\beta(X - \bar{X}_2 - \epsilon) + 2\alpha\beta(X - \bar{X}_1 - \epsilon)(X - \bar{X}_2 - \epsilon) + \beta^2(X - \bar{X}_2 - \epsilon)^2 \right) \gamma(\theta) d\theta.\end{aligned}$$

The first-order condition (as before, multiplied by 1/2) is given by:

$$\begin{aligned}0 &= \frac{d\mathcal{L}_1}{dX} + \int_{\theta \in \Theta_2} (\beta(\hat{S} - \bar{S}) + 2\alpha\beta X^{***} - \alpha\beta(\bar{X}_1 + \bar{X}_2) - 2\alpha\beta\epsilon + \beta^2(X^{***} - \bar{X}_2 - \epsilon)) \gamma(\theta) d\theta \\ &= \frac{d\mathcal{L}_1}{dX} + \int_{\theta \in \Theta_2} \left((2\alpha\beta + \beta^2)X^{***} + \beta(\hat{S} - \bar{S}) - \alpha\beta(\bar{X}_1 + \bar{X}_2) - (2\alpha\beta + \beta^2)\epsilon - \beta^2\bar{X}_2 \right) \gamma(\theta) d\theta \\ &= \frac{d\mathcal{L}_1}{dX} + (2\alpha\beta + \beta^2)(1 - G(\bar{\epsilon}_2))X^{***} \\ &\quad + \left(\beta(\hat{S} - \bar{S}) - \alpha\beta(\bar{X}_1 + \bar{X}_2) - \beta^2\bar{X}_2 \right) (1 - G(\bar{\epsilon}_2)) - (2\alpha\beta + \beta^2) \int_{\bar{\epsilon}_2}^{\infty} \epsilon g(\epsilon) d\epsilon,\end{aligned}$$

where $\frac{d\mathcal{L}_1}{dX}$ is calculated at the optimum, X^{***} . Using that $\int_{\bar{\epsilon}_2}^{\infty} \epsilon g(\epsilon) d\epsilon = \mathcal{G}(\bar{\epsilon}_2) - \bar{\epsilon}_2 G(\bar{\epsilon}_2)$ and $\bar{\epsilon}_2 = X^{***} - \bar{X}_2$, we get

$$\begin{aligned}&(2\alpha\beta + \beta^2)(1 - G(\bar{\epsilon}_2))X^{***} + \left(\beta(\hat{S} - \bar{S}) - \alpha\beta(\bar{X}_1 + \bar{X}_2) - \beta^2\bar{X}_2 \right) (1 - G(\bar{\epsilon}_2)) + (2\alpha\beta + \beta^2)(\bar{\epsilon}_2 G(\bar{\epsilon}_2) - \mathcal{G}(\bar{\epsilon}_2)) \\ &= (2\alpha\beta + \beta^2)(X^{***} - \bar{X}_2) + \left(\beta(\hat{S} - \bar{S}) - \alpha\beta(\bar{X}_1 - \bar{X}_2) \right) (1 - G(\bar{\epsilon}_2)) - (2\alpha\beta + \beta^2)\mathcal{G}(\bar{\epsilon}_2).\end{aligned}$$

Taken together, we get the combined first-order condition

$$\begin{aligned}&(\alpha^2 + \lambda)(X^{***} - \bar{X}_1) + \alpha(1 - G(\bar{\epsilon}_1))(\hat{S} - \bar{S}) + \lambda(\bar{X}_1 - \hat{X}) - \alpha^2\mathcal{G}(\bar{\epsilon}_1) + \\ &+ (2\alpha\beta + \beta^2)(X^{***} - \bar{X}_2) + \beta(1 - G(\bar{\epsilon}_2))(\hat{S} - \bar{S}) - \alpha\beta(1 - G(\bar{\epsilon}_2))(\bar{X}_1 - \bar{X}_2) - (2\alpha\beta + \beta^2)\mathcal{G}(\bar{\epsilon}_2) = 0,\end{aligned}$$

which simplifies to

$$\begin{aligned}X^{***} &= \frac{\alpha^2 + \lambda}{(\alpha + \beta)^2 + \lambda} X^{**} + \\ &\quad + \frac{\beta(1 - G(\bar{\epsilon}_2))}{((\alpha + \beta)^2 + \lambda)} (\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) - \hat{S}) + \frac{2\alpha\beta + \beta^2}{((\alpha + \beta)^2 + \lambda)} (\bar{X}_2 + \mathcal{G}(\bar{\epsilon}_2)),\end{aligned}$$

where $\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) > \hat{S}$ is the spread at $X = \bar{X}_2$ with $\epsilon = 0$ and $\nu = 0$. To show that $X^{***} > X^{**}$, we use that $\mathcal{G}(\bar{\epsilon}_2) \geq \bar{\epsilon}_2 = X^{***} - \bar{X}_2$. Consequently, $\mathcal{G}(\bar{\epsilon}_2) + \bar{X}_2 \geq X^{***}$, and thus the last term on the right-hand side can be bounded from below:

$$\begin{aligned} X^{***} \geq & \frac{\alpha^2 + \lambda}{(\alpha + \beta)^2 + \lambda} X^{**} + \\ & + \frac{\beta(1 - G(\bar{\epsilon}_2))}{(\alpha + \beta)^2 + \lambda} (\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) - \hat{S}) + \frac{2\alpha\beta + \beta^2}{(\alpha + \beta)^2 + \lambda} X^{***}. \end{aligned}$$

Solving for X^{***} results in

$$X^{***} \geq X^{**} + \frac{\beta(1 - G(\bar{\epsilon}_2))}{\alpha^2 + \lambda} (\bar{S} + \alpha(\bar{X}_1 - \bar{X}_2) - \hat{S}).$$

B Proofs of Section 4 (with Lending Facility)

In this appendix, we report the proofs of the optimal supply of reserves under uncertainty when a central bank lending facility is present.

B.1 Mean spread and variance of the spread with a lending facility

The expected spread in the presence of the lending facility described in Section 2.2 can be written as follows

$$\begin{aligned}
\mathbb{E}[S] &= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S} + \nu) f(\nu) g(\epsilon) d\nu d\epsilon + \\
&\quad \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \int_{-\infty}^{\infty} [\bar{S} + \nu - \alpha (X - \bar{X} - \epsilon)] f(\nu) g(\epsilon) d\nu d\epsilon + \\
&\quad \int_{\bar{\epsilon}_L}^{\infty} \int_{-\infty}^{\infty} (\bar{S}_L + \nu) f(\nu) g(\epsilon) d\nu d\epsilon \\
&= \bar{S} G(\bar{\epsilon}) + \bar{S} [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] + \bar{S}_L [1 - G(\bar{\epsilon}_L)] - \alpha \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} (X - \bar{X} - \epsilon) g(\epsilon) d\epsilon \\
&= \bar{S} + (\bar{S}_L - \bar{S}) [1 - G(\bar{\epsilon}_L)] - \alpha \left\{ \bar{\epsilon} [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon g(\epsilon) d\epsilon \right\},
\end{aligned}$$

where the first equality holds because ϵ and ν are independent, the second equality holds because ν has zero mean, and the third equality uses $\bar{\epsilon} = X - \bar{X}$ by definition. Using integration by parts, we can write

$$\int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon g(\epsilon) d\epsilon = \bar{\epsilon}_L G(\bar{\epsilon}_L) - \bar{\epsilon} G(\bar{\epsilon}) - \int_{-\infty}^{\bar{\epsilon}_L} G(\epsilon) d\epsilon + \int_{-\infty}^{\bar{\epsilon}} G(\epsilon) d\epsilon = \bar{\epsilon}_L G(\bar{\epsilon}_L) - \bar{\epsilon} G(\bar{\epsilon}) - [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})]$$

Substituting this expression in the last term of the previous formula, we can write the expected spread as

$$\begin{aligned}
\mathbb{E}[S] &= \bar{S} + (\bar{S}_L - \bar{S}) [1 - G(\bar{\epsilon}_L)] - \alpha \{ \bar{\epsilon} G(\bar{\epsilon}_L) - \bar{\epsilon} G(\bar{\epsilon}) - \bar{\epsilon}_L G(\bar{\epsilon}_L) + \bar{\epsilon} G(\bar{\epsilon}) + [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] \} \\
&= \bar{S} + (\bar{S}_L - \bar{S}) [1 - G(\bar{\epsilon}_L)] - \alpha \left[-\frac{\bar{S}_L - \bar{S}}{\alpha} G(\bar{\epsilon}_L) + \mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon}) \right] \\
&= \bar{S}_L - \alpha [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})],
\end{aligned}$$

where the second equality follows from $\bar{\epsilon}_L = \bar{\epsilon} + \frac{\bar{S}_L - \bar{S}}{\alpha}$.

From the equation above, it follows that: (i) $\mathbb{E}[S] \leq \bar{S}_L$ for all X because $\mathcal{G}(x)$ is increasing everywhere, which implies that $\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon}) \geq 0$ for all X ; (ii) $\mathbb{E}[S] \geq \bar{S}$ for all X because the function $\mathcal{G}(x) - x$ is decreasing everywhere, which implies that $\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon}) \leq \bar{\epsilon}_L - \bar{\epsilon}$; (iii) $\frac{d\mathbb{E}[S]}{d\bar{S}_L} = 1 - G(\bar{\epsilon}_L) \geq 0$ for all $\bar{S}_L$ and X , which means that $\mathbb{E}[S]$ is increasing with $\bar{S}_L$; (iv) $\frac{d\mathbb{E}[S]}{dX} = -\alpha [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] \leq 0$ for all $\bar{S}_L$ and X , which means that $\mathbb{E}[S]$ is decreasing with X ; (v) $\lim_{X \rightarrow -\infty} \mathbb{E}[S] = \bar{S}_L$ because both $\mathcal{G}$ functions go to zero as X goes to minus infinity; and (vi) $\lim_{X \rightarrow \infty} \mathbb{E}[S] = \bar{S}$ because $\lim_{x \rightarrow \infty} (\mathcal{G}(x+d) - \mathcal{G}(x)) = \lim_{x \rightarrow \infty} \left\{ (x+d)G(x+d) - xG(x) - \int_{-\infty}^{x+d} u g(u) du + \int_{-\infty}^x u g(u) du \right\} = d$ for all x and d , where we used integration by parts and the fact that ϵ has zero mean (so that the last two integrals go to zero in the limit) and finite variance (so that $x[G(x+d) - G(x)]$ goes to zero in the limit). Moreover, since $\mathbb{E}[S]$ is a decreasing function of X , the limit for $X \rightarrow -\infty$ is approached from below and the limit for $X \rightarrow \infty$ is approached from above.

Since $\frac{d\mathbb{E}[S]}{dX} = -\alpha [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] \geq -\alpha$ for all $\bar{S}_L$ and X , and since the deterministic spread is

$$S^{det} = \begin{cases} \bar{S}_L & \text{if } X \leq \bar{X} - \frac{\bar{S}_L - \bar{S}}{\alpha}, \\ \bar{S} - \alpha(X - \bar{X}) & \text{if } \bar{X} - \frac{\bar{S}_L - \bar{S}}{\alpha} < X < \bar{X}, \\ \bar{S} & \text{if } X \geq \bar{X}, \end{cases}$$

there exists a unique level of the ex-ante supply of reserves $\tilde{X} \in \left(\bar{X} - \frac{\bar{S}_L - \bar{S}}{\alpha}, \bar{X} \right)$ such that the expected spread is equal to the deterministic one, with $\mathbb{E}[S] \leq S^{det}$ for all $X < \tilde{X}$ and $\mathbb{E}[S] \geq S^{det}$ for all $X > \tilde{X}$. If the distribution of horizontal shocks $g(\epsilon)$ has infinite support, the inequalities hold in the strict sense; obviously, if $g(\epsilon)$ has finite support, the expected spread is exactly equal to the deterministic one for sufficiently low and for sufficiently large values of X . Finally, if $g(\epsilon)$ is symmetric around zero, $G(u) + G(-u) = 1$

for all u and therefore $\frac{d}{du} [\mathcal{G}(u) - \mathcal{G}(-u)] = G(u) + G(-u) = 1$ for all u ; this implies $[\mathcal{G}(u) - \mathcal{G}(-u)] = u$ because $\mathcal{G}(0) - \mathcal{G}(-0) = 0$. As a result, $\mathbb{E} \left[S \left(X = \bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha} \right) \right] = \bar{S}_L - \alpha \left(\bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha} - \bar{X} + \frac{\bar{S}_L - \bar{S}}{\alpha} \right) = \frac{\bar{S}_L + \bar{S}}{2} = S^{det} \left(X = \bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha} \right)$, which proves that if $g(\epsilon)$ is symmetric around zero, the expected spread intersects the deterministic one at the midpoint between the kinks in the reserve demand curve, i.e., $\bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha}$.

To derive the variance of the spread in the presence of the lending facility, we first derive the expected value of the squared spread as follows

$$\begin{aligned}
\mathbb{E}[S^2] &= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S} + v)^2 f(v) g(\epsilon) dv d\epsilon + \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \int_{-\infty}^{\infty} [\bar{S} + v - \alpha(X - \bar{X} - \epsilon)]^2 f(v) g(\epsilon) dv d\epsilon + \\
&\quad \int_{\bar{\epsilon}_L}^{\infty} \int_{-\infty}^{\infty} (\bar{S}_L + v)^2 f(v) g(\epsilon) dv d\epsilon \\
&= \int_{-\infty}^{\bar{\epsilon}} \int_{-\infty}^{\infty} (\bar{S}^2 + 2v\bar{S} + v^2) f(v) g(\epsilon) dv d\epsilon + \\
&\quad \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \int_{-\infty}^{\infty} [(\bar{S} + v)^2 - 2\alpha(\bar{S} + v)(\bar{\epsilon} - \epsilon) + \alpha^2(\bar{\epsilon} - \epsilon)^2] f(v) g(\epsilon) dv d\epsilon + \\
&\quad \int_{\bar{\epsilon}_L}^{\infty} \int_{-\infty}^{\infty} (\bar{S}_L^2 + 2v\bar{S}_L + v^2) f(v) g(\epsilon) dv d\epsilon \\
&= \bar{S}^2 G(\bar{\epsilon}) + 2\bar{S} \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} v f(v) dv \right) g(\epsilon) d\epsilon + \sigma_v^2 G(\bar{\epsilon}) + \bar{S}_L^2 [1 - G(\bar{\epsilon}_L)] + \\
&\quad 2\bar{S}_L \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} v f(v) dv \right) g(\epsilon) d\epsilon + \sigma_v^2 [1 - G(\bar{\epsilon}_L)] + \bar{S}^2 [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] + \\
&\quad 2\bar{S} \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} v f(v) dv \right) g(\epsilon) d\epsilon + \sigma_v^2 [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - 2\alpha\bar{S} \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} f(v) dv \right) (\bar{\epsilon} - \epsilon) g(\epsilon) d\epsilon - \\
&\quad 2\alpha \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} v f(v) dv \right) (\bar{\epsilon} - \epsilon) g(\epsilon) d\epsilon + \alpha^2 \int_{-\infty}^{\bar{\epsilon}} \left(\int_{-\infty}^{\infty} f(v) dv \right) (\bar{\epsilon} - \epsilon)^2 g(\epsilon) d\epsilon \\
&= \sigma_v^2 + \bar{S}_L^2 [1 - G(\bar{\epsilon}_L)] + \bar{S}^2 G(\bar{\epsilon}_L) - 2\alpha\bar{S} \left\{ \bar{\epsilon} [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon g(\epsilon) d\epsilon \right\} + \\
&\quad \alpha^2 \left\{ \bar{\epsilon}^2 [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - 2\bar{\epsilon} \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon g(\epsilon) d\epsilon + \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon^2 g(\epsilon) d\epsilon \right\},
\end{aligned}$$

where the first equality holds because ϵ and v are independent, and the fourth equality

holds because ν has zero mean.

Using integration by parts, the last integral in the equation above can be written as

$$\begin{aligned} \int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon^2 g(\epsilon) d\epsilon &= \int_{-\infty}^{\bar{\epsilon}_L} \epsilon^2 g(\epsilon) d\epsilon - \int_{-\infty}^{\bar{\epsilon}} \epsilon^2 g(\epsilon) d\epsilon \\ &= (\bar{\epsilon}_L)^2 G(\bar{\epsilon}_L) - 2\bar{\epsilon}_L \mathcal{G}(\bar{\epsilon}_L) + 2\mathbb{G}(\bar{\epsilon}_L) - \bar{\epsilon}^2 G(\bar{\epsilon}) + 2\bar{\epsilon} \mathcal{G}(\bar{\epsilon}) - 2\mathbb{G}(\bar{\epsilon}), \end{aligned}$$

where $\mathbb{G}(x) = \int_{-\infty}^x \mathcal{G}(u) du$ is the super-super cumulative distribution function. Using this expression and the expression for $\int_{\bar{\epsilon}}^{\bar{\epsilon}_L} \epsilon g(\epsilon) d\epsilon$ derived above, we can write

$$\begin{aligned} \mathbb{E}[S^2] &= \sigma_v^2 + \bar{S}_L^2 [1 - G(\bar{\epsilon}_L)] + \bar{S}^2 G(\bar{\epsilon}_L) - \\ &\quad 2\alpha \bar{S} \{ \bar{\epsilon} [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - \bar{\epsilon}_L G(\bar{\epsilon}_L) + \bar{\epsilon} G(\bar{\epsilon}) + \mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon}) \} + \\ &\quad \alpha^2 \{ \bar{\epsilon}^2 [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] - 2\bar{\epsilon} [\bar{\epsilon}_L G(\bar{\epsilon}_L) - \bar{\epsilon} G(\bar{\epsilon}) - \mathcal{G}(\bar{\epsilon}_L) + \mathcal{G}(\bar{\epsilon})] + \\ &\quad (\bar{\epsilon}_L)^2 G(\bar{\epsilon}_L) - \bar{\epsilon}^2 G(\bar{\epsilon}) - 2\bar{\epsilon}_L \mathcal{G}(\bar{\epsilon}_L) + 2\bar{\epsilon} \mathcal{G}(\bar{\epsilon}) + 2\mathbb{G}(\bar{\epsilon}_L) - 2\mathbb{G}(\bar{\epsilon}) \} \\ &= \sigma_v^2 + \bar{S}_L^2 [1 - G(\bar{\epsilon}_L)] + \bar{S}^2 G(\bar{\epsilon}_L) + 2\alpha \bar{S} \{ (\bar{\epsilon}_L - \bar{\epsilon}) G(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon}_L) + \mathcal{G}(\bar{\epsilon}) \} + \\ &\quad \alpha^2 \{ (\bar{\epsilon}_L - \bar{\epsilon})^2 G(\bar{\epsilon}_L) - 2(\bar{\epsilon}_L - \bar{\epsilon}) \mathcal{G}(\bar{\epsilon}) + 2[\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})] \} \\ &= \sigma_v^2 + \bar{S}_L^2 [1 - G(\bar{\epsilon}_L)] + \bar{S}^2 G(\bar{\epsilon}_L) + 2\bar{S} (\bar{S}_L - \bar{S}) G(\bar{\epsilon}_L) - 2\alpha \bar{S} [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] + \\ &\quad (\bar{S}_L - \bar{S})^2 G(\bar{\epsilon}_L) - 2\alpha (\bar{S}_L - \bar{S}) \mathcal{G}(\bar{\epsilon}) + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})] \\ &= \sigma_v^2 + \bar{S}_L^2 - 2\alpha [\bar{S}_L \mathcal{G}(\bar{\epsilon}_L) - \bar{S} \mathcal{G}(\bar{\epsilon})] + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})], \end{aligned}$$

where the third equality uses $\bar{\epsilon}_L = \bar{\epsilon} + \frac{\bar{S}_L - \bar{S}}{\alpha}$. Putting together this expression with the one for the expected spread, we obtain the variance of the spread as

$$\begin{aligned} \text{Var}[S] &= \mathbb{E}[S^2] - (\mathbb{E}[S])^2 = \sigma_v^2 + \bar{S}_L^2 - 2\alpha [\bar{S}_L \mathcal{G}(\bar{\epsilon}_L) - \bar{S} \mathcal{G}(\bar{\epsilon})] + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})] - \\ &\quad \bar{S}_L^2 + 2\alpha \bar{S}_L [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] - \alpha^2 [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})]^2 \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 - 2\alpha (\bar{S}_L - \bar{S}) \mathcal{G}(\bar{\epsilon}_L) - \alpha^2 \{ [\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}] \}^2 + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})]. \end{aligned}$$

To prove the limiting behavior of $\text{Var}[S(X)]$ as X goes to plus and minus infinity, we use the following equalities, which are derived using integration by parts:

$$\begin{aligned}\mathcal{G}(x) &= \int_{-\infty}^x G(u)du = xG(x) - \int_{-\infty}^x u g(u)du, \\ \mathbb{G}(x) &= \int_{-\infty}^x \mathcal{G}(u)du = x\mathcal{G}(x) - \frac{x^2}{2}G(x) + \int_{-\infty}^x \frac{u^2}{2}g(u)du.\end{aligned}$$

Plugging these equalities in the expression for the variance above, we obtain:

$$\begin{aligned}\lim_{X \rightarrow \infty} \text{Var}[S] &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 + \\ &\quad \lim_{X \rightarrow \infty} \left\{ -2\alpha (\bar{S}_L - \bar{S}) \mathcal{G}(\bar{\epsilon}_L) - \alpha^2 \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\}^2 + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})] \right\} \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 + \\ &\quad \lim_{X \rightarrow \infty} \left\{ -2\alpha (\bar{S}_L - \bar{S}) \bar{\epsilon}_L G(\bar{\epsilon}_L) + 2\alpha^2 \left[\bar{\epsilon}_L \mathcal{G}(\bar{\epsilon}_L) - \frac{(\bar{\epsilon}_L)^2}{2} G(\bar{\epsilon}_L) - \bar{\epsilon} \mathcal{G}(\bar{\epsilon}) + \frac{\bar{\epsilon}^2}{2} G(\bar{\epsilon}) \right] \right\} \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 + \lim_{X \rightarrow \infty} \left\{ -2\alpha (\bar{S}_L - \bar{S}) \bar{\epsilon}_L G(\bar{\epsilon}_L) + 2\alpha^2 \left[\frac{(\bar{\epsilon}_L)^2}{2} G(\bar{\epsilon}_L) - \frac{\bar{\epsilon}^2}{2} G(\bar{\epsilon}) \right] \right\} \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 + \lim_{X \rightarrow \infty} \left\{ -2\alpha (\bar{S}_L - \bar{S}) \bar{\epsilon}_L + \alpha^2 [(\bar{\epsilon}_L)^2 - \bar{\epsilon}^2] \right\} \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 - 2(\bar{S}_L - \bar{S})^2 + (\bar{S}_L - \bar{S})^2 = \sigma_v^2,\end{aligned}$$

where we used $\lim_{x \rightarrow \infty} G(x) = 1$, the fact that ϵ has zero mean and finite variance, and that $\bar{\epsilon}_L = \bar{\epsilon} + \frac{\bar{S}_L - \bar{S}}{\alpha}$. Similarly,

$$\begin{aligned}\lim_{X \rightarrow -\infty} \text{Var}[S] &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 + \\ &\quad \lim_{X \rightarrow -\infty} \left\{ -2\alpha (\bar{S}_L - \bar{S}) \mathcal{G}(\bar{\epsilon}_L) - \alpha^2 \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\}^2 + 2\alpha^2 [\mathbb{G}(\bar{\epsilon}_L) - \mathbb{G}(\bar{\epsilon})] \right\} \\ &= \sigma_v^2 + (\bar{S}_L - \bar{S})^2 - \alpha^2 \lim_{X \rightarrow -\infty} (\bar{\epsilon}_L - \bar{\epsilon})^2 = \sigma_v^2,\end{aligned}$$

where we used the equalities derived above and the fact that $\lim_{x \rightarrow -\infty} xG(x) = \lim_{x \rightarrow -\infty} x^2G(x) =$

$\lim_{x \rightarrow -\infty} x\mathcal{G}(x) = 0$ because ϵ has finite variance.

The derivative of $\text{Var}[S]$ with respect to $\bar{S}_L$ is

$$\begin{aligned} \frac{d\text{Var}[S]}{d\bar{S}_L} &= 2(\bar{S}_L - \bar{S}) - 2\alpha\mathcal{G}(\bar{\epsilon}_L) - 2\alpha(\bar{S}_L - \bar{S}) \frac{G(\bar{\epsilon}_L)}{\alpha} \\ &\quad - 2\alpha^2 \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\} \frac{G(\bar{\epsilon}_L) - 1}{\alpha} + 2\alpha^2 \frac{\mathcal{G}(\bar{\epsilon}_L)}{\alpha} \\ &= 2(\bar{S}_L - \bar{S}) [1 - G(\bar{\epsilon}_L)] + 2\alpha \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\} [1 - G(\bar{\epsilon}_L)] \\ &= 2\alpha [1 - G(\bar{\epsilon}_L)] [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] \geq 0, \end{aligned}$$

where we used $\bar{\epsilon}_L = \bar{\epsilon} + \frac{\bar{S}_L - \bar{S}}{\alpha}$, and the last inequality follows from the fact that $\mathcal{G}(x)$ is increasing everywhere. This result proves that the variance of the spread increases with the lending facility spread.

The derivative of $\text{Var}[S]$ with respect to X is

$$\begin{aligned} \frac{d\text{Var}[S]}{dX} &= -2\alpha(\bar{S}_L - \bar{S}) G(\bar{\epsilon}_L) - 2\alpha^2 \{[\mathcal{G}(\bar{\epsilon}_L) - \bar{\epsilon}_L] - [\mathcal{G}(\bar{\epsilon}) - \bar{\epsilon}]\} [G(\bar{\epsilon}_L) - G(\bar{\epsilon})] \\ &\quad + 2\alpha^2 [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] \\ &= -2\alpha(\bar{S}_L - \bar{S}) G(\bar{\epsilon}) + 2\alpha^2 [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] [1 - G(\bar{\epsilon}_L) + G(\bar{\epsilon})] \\ &= 2\alpha^2 \left\{ [\mathcal{G}(\bar{\epsilon}_L) - \mathcal{G}(\bar{\epsilon})] [1 - G(\bar{\epsilon}_L) + G(\bar{\epsilon})] - \frac{\bar{S}_L - \bar{S}}{\alpha} G(\bar{\epsilon}) \right\} \geq 0, \end{aligned}$$

which shows that the variance of the spread is non-monotonic with X .

To prove that if the g is symmetric around zero, $\text{Var}[S]$ is an even function of X around $X = \bar{X} - \frac{\bar{S}_L - \bar{S}}{2\alpha}$ (i.e., around $\bar{\epsilon} = -\frac{\bar{S}_L - \bar{S}}{2\alpha}$), we first show that its derivative with respect to X is an odd function. Let $d \equiv \frac{\bar{S}_L - \bar{S}}{2\alpha}$, so that $\bar{\epsilon}_L = \bar{\epsilon} + 2d$, and $H(\bar{\epsilon}) \equiv (2\alpha^2)^{-1} \frac{d\text{Var}[S]}{dX}$.

Then, for any $z > 0$,

$$H(-d + z) = [\mathcal{G}(d + z) - \mathcal{G}(-d + z)] [1 - G(d + z) + G(-d + z)] - 2dG(-d + z), \text{ and}$$

$$H(-d - z) = [\mathcal{G}(d - z) - \mathcal{G}(-d - z)] [1 - G(d - z) + G(-d - z)] - 2dG(-d - z).$$

Since $g(\epsilon)$ is symmetric around zero, $G(-u) = 1 - G(u)$ and therefore $\frac{d}{du} [\mathcal{G}(u) - \mathcal{G}(-u)] = G(u) + G(-u) = 1$, which implies $[\mathcal{G}(u) - \mathcal{G}(-u)] = u$ because $\mathcal{G}(0) - \mathcal{G}(-0) = 0$. Using these results, we can show

$$\begin{aligned} H(-d + z) + H(-d - z) &= [\mathcal{G}(d + z) - \mathcal{G}(-d + z)] [1 - G(d + z) + G(-d + z)] - 2dG(-d + z) + \\ &\quad [\mathcal{G}(d - z) - \mathcal{G}(-d - z)] [1 - G(d - z) + G(-d - z)] - 2dG(-d - z) \\ &= [\mathcal{G}(d + z) - \mathcal{G}(-d + z)] [G(-d - z) + G(-d + z)] - 2dG(-d + z) + \\ &\quad [\mathcal{G}(d - z) - \mathcal{G}(-d - z)] [G(-d + z) + G(-d - z)] - 2dG(-d - z) \\ &= [\mathcal{G}(d + z) - \mathcal{G}(-d + z) + \mathcal{G}(d - z) - \mathcal{G}(-d - z)] [G(-d - z) + G(-d + z)] - \\ &\quad 2d [G(-d + z) + G(-d - z)] \\ &= [d + z + d - z] [G(-d - z) + G(-d + z)] - 2d [G(-d + z) + G(-d - z)] = 0, \end{aligned}$$

which implies that $H(\bar{\epsilon})$ is an odd function around $\bar{\epsilon} = -d$. As a result, $\frac{d\text{Var}[S]}{dX}$ is an odd function around $\bar{\epsilon} = -d$, which implies that $\text{Var}[S]$ is an even function around $\bar{\epsilon} = -d$.

To prove that if g is also log-concave, $\bar{\epsilon} = -d$ is a local maximum of the variance, we first observe that $\left. \frac{d\text{Var}[S]}{dX} \right|_{\bar{\epsilon}=-d} = 0$ because $\frac{d\text{Var}[S]}{dX}$ is odd around $\bar{\epsilon} = -d$. Then, we calculate the second derivative of the variance at $\bar{\epsilon} = -d$:

$$\begin{aligned} \left. \frac{d^2\text{Var}[S]}{dX^2} \right|_{\bar{\epsilon}=-d} &= 2\alpha^2 \{ [G(d) - G(-d)] [1 - G(d) + G(-d)] + [\mathcal{G}(d) - \mathcal{G}(-d)] [g(d) - g(-d)] - 2dg(-d) \} \\ &= 2\alpha^2 \{ [2dg(z)] [2G(-d)] - 2dg(-d) \} = 4\alpha^2 dG(-d) \left[\frac{g(z)}{1/2} - \frac{g(-d)}{G(-d)} \right], \end{aligned}$$

with $z \in (-d, d)$; the second equality uses the mean value theorem on $G(d) - G(-d)$ and the fact that g is symmetric around zero, which implies $1 - G(d) = G(-d)$ and $g(-d) = g(d)$. Since g is symmetric around zero and log-concave, $g(z) < g(0)$ for all z , $G(0) = 1/2$, and the reverse hazard rate, $\frac{g(x)}{G(x)}$, is decreasing. As a result,

$$\left. \frac{d^2 \text{Var}[S]}{dX^2} \right|_{\bar{\epsilon} = -d} = 4\alpha^2 d G(-d) \left[\frac{g(z)}{1/2} - \frac{g(-d)}{G(-d)} \right] < 4\alpha^2 d G(-d) \left[\frac{g(0)}{G(0)} - \frac{g(-d)}{G(-d)} \right] < 0.$$

Combined with $\left. \frac{d \text{Var}[S]}{dX} \right|_{\bar{\epsilon} = -d} = 0$, this inequality proves that $\bar{\epsilon} = -d$ is a local maximum of $\text{Var}[S]$.

B.2 Optimal policy with the lending facility

B.2.1 Exogenous lending facility spread

The loss function to minimize is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}^{\text{No LF}} + \frac{1}{2} \int_{\bar{\epsilon}^{\text{SRF}}}^{\infty} \left(\bar{S}^L - \hat{S} \right)^2 g(\epsilon) d\epsilon - \frac{1}{2} \int_{\bar{\epsilon}^{\text{SRF}}}^{\infty} \left(\bar{S} - \alpha(X - \bar{X} - \epsilon) - \hat{S} \right)^2 g(\epsilon) d\epsilon + \\ & \lambda^L \int_{\bar{\epsilon}^{\text{SRF}}}^{\infty} \left(-\frac{\bar{S}^L - \bar{S}}{\alpha} - (X - \bar{X} - \epsilon) \right) g(\epsilon) d\epsilon, \end{aligned}$$

where $\mathcal{L}^{\text{No LF}}$ is the central bank loss function absent the lending facility (see Section 3).

The first-order condition with respect to X is

$$0 = \frac{\partial \mathcal{L}^{\text{No LF}}}{\partial X} + \alpha \left(1 - G(\bar{\epsilon}^{\text{SRF}}) \right) (\bar{S}^L - \hat{S}) - \lambda^L \left(1 - G(\bar{\epsilon}^{\text{SRF}}) \right) + \alpha^2 \left(\mathcal{Z}(\bar{\epsilon}^{\text{SRF}}) - \bar{\epsilon}^{\text{SRF}} \right),$$

which implies

$$\frac{\partial \mathcal{L}^{\text{No LF}}}{\partial X} = -\alpha \left(1 - G(\bar{\epsilon}^{\text{SRF}}) \right) (\bar{S}^L - \hat{S}) - \alpha^2 \left(\mathcal{Z}(\bar{\epsilon}^{\text{SRF}}) - \bar{\epsilon}^{\text{SRF}} \right) + \lambda^L \left(1 - G(\bar{\epsilon}^{\text{SRF}}) \right) \geq 0.$$

That is, the optimal supply in the presence of a lending facility with exogenous lending rate can be either greater or smaller than the optimal supply absent the facility (see Proposition 2 of the paper). Also, from the equation above, it is clear that if λ^L is sufficiently small, the right-hand side of the equation is negative; since the loss function is convex at the optimum (i.e., its first derivative is an increasing function of X), this implies that the optimal supply of reserves with the facility is lower than the optimal supply with the facility: $\frac{\partial \mathcal{L}^{\text{No LF}}(X^{B*})}{\partial X} < 0 = \frac{\partial \mathcal{L}^{\text{No LF}}(X^{**})}{\partial X}$, where X^{B*} and X^{**} are the optima with and without facility.

B.2.2 Endogenous lending facility spread

The first-order condition with respect to $\bar{S}^L$ is

$$0 = \bar{S}^L - \hat{S} - \frac{\lambda^L}{\alpha}$$

Plugging this equation into the first-order condition for reserves, we obtain

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}^{\text{No LF}}}{\partial X} + \alpha \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right) (\bar{S}^L - \hat{S}) - \lambda^L \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right) + \alpha^2 \left(\mathcal{G}(\bar{\epsilon}^{\text{SRF}}) - \bar{\epsilon}^{\text{SRF}}\right) \\ &= \frac{\partial \mathcal{L}^{\text{No LF}}}{\partial X} + \alpha^2 \left(\mathcal{G}(\bar{\epsilon}^{\text{SRF}}) - \bar{\epsilon}^{\text{SRF}}\right); \end{aligned}$$

this equation implies that if the central bank chooses the lending rate spread of its lending facility optimally, optimal reserves are smaller than without the facility; that is, $\frac{\partial \mathcal{L}^{\text{No LF}}(X^{L*})}{\partial X} < 0 = \frac{\partial \mathcal{L}^{\text{No LF}}(X^{**})}{\partial X}$, where X^{L*} and X^{**} are the optima with and without facility.

The second-order derivatives are

$$\begin{aligned}\frac{\partial^2 \mathcal{L}}{\partial (\bar{S}^L)^2} &= \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right) - \frac{g(\bar{\epsilon}^{\text{SRF}})}{\alpha} \left(\bar{S}^L - \hat{S} - \frac{\lambda^L}{\alpha}\right), \\ \frac{\partial^2 \mathcal{L}}{\partial (\bar{S}^L)^2} \Big|_{x^*, \bar{S}^{L*}} &= \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right) > 0;\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 \mathcal{L}}{\partial X^2} &= \frac{\partial^2 \mathcal{L}^{\text{No LF}}}{\partial X^2} - \frac{g(\bar{\epsilon}^{\text{SRF}})}{\alpha} \left(\bar{S}^L - \hat{S} - \frac{\lambda^L}{\alpha}\right) - \alpha^2 \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right), \\ \frac{\partial^2 \mathcal{L}}{\partial X^2} \Big|_{x^*, \bar{S}^{L*}} &= \frac{\partial^2 \mathcal{L}^{\text{No LF}}}{\partial X^2} \Big|_{x^*, \bar{S}^{L*}} - \alpha^2 \left(1 - G(\bar{\epsilon}^{\text{SRF}})\right); \end{aligned}$$

$$\begin{aligned}\frac{\partial^2 \mathcal{L}}{\partial X \partial \bar{S}^L} &= -\frac{g(\bar{\epsilon}^{\text{SRF}})}{\alpha} \left(\bar{S}^L - \hat{S} - \frac{\lambda^L}{\alpha}\right) \\ \frac{\partial^2 \mathcal{L}}{\partial X \partial \bar{S}^L} \Big|_{x^*, \bar{S}^{L*}} &= 0.\end{aligned}$$

The second-order condition is therefore simply

$$SOC(X) := \frac{\partial^2 \mathcal{L}}{\partial X^2} \Big|_{x^*, \bar{S}^{L*}} \geq 0$$

From these results, we can derive the following comparative statics with respect to the facility-usage penalty λ^L :

$$\frac{d\bar{S}^L}{d\lambda^L} = \frac{1}{\alpha} > 0$$

$$\frac{dX}{d\lambda^L} = \frac{1 - G(\bar{\epsilon}^{\text{SRF}})}{SOC(X)} \geq 0$$

That is, both the optimal lending facility rate and the optimal reserve supply increase with

the penalty cost of facility usage. The intuition is simple: if the cost is higher, the central bank want to increase the facility lending rate to reduce its usage; since the cap on money market rates is now higher, the central bank wants to provide more reserves to reduce rate volatility.

C Empirical evidence

In this appendix, we show empirical evidence that motivates our model of the reserve demand curve, using data from the United States. Panel (a) of Figure C.1 provides a visualization of the “realized” demand for reserves in the U.S. from 2010 to June 2025. The figure shows the spread between the federal funds rate and the interest rate on reserve balances against the level of aggregate reserves normalized by total assets in the banking sector. Note that this figure shows equilibrium realizations over time, not the underlying demand schedule. With that caveat in mind, for each of the different sub-periods, the figure shows a nonlinear, downward-sloping relationship between prices and quantities that flattens at high levels of reserves, consistent with the demand curve in equation (2).

Importantly, Panel (a) also shows that this relationship has shifted over time, both vertically and horizontally. Vertical and horizontal shifts are captured in our model by the shocks ν and ϵ respectively. Vertical shifts of the demand for reserves are driven, for instance, by changes in the costs that banks face when expanding their balance sheets through borrowing: if these costs increase, the rate at which banks are willing to borrow decreases, pushing the reserve demand curve downward (Afonso et al., 2019). An example of these balance-sheet costs is the Supplementary Leverage Ratio (SLR), which sets a minimum requirement for the ratio of bank equity over assets (Duffie, 2018). Market innovations that change banks’ ability to net their lending and borrowing positions, such as the Sponsored Repo Service operated by the Fixed Income Clearing Corporation, also

affect bank balance-sheet costs by reducing balance-sheet size (Afonso et al., 2021).

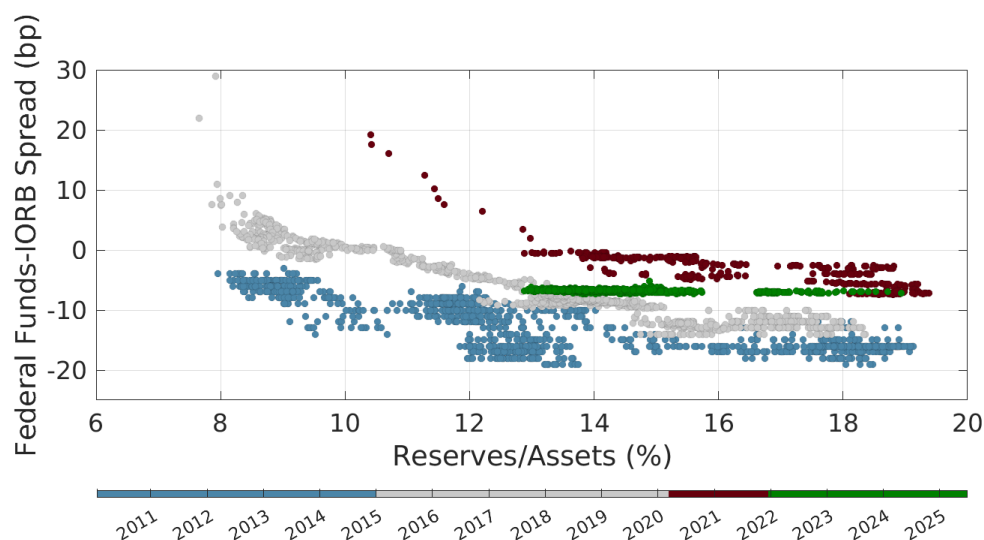
Horizontal shifts can be due to regulatory and supervisory measures that affect banks' demand for reserves for every price level. Examples are the Liquidity Coverage Ratio, which requires banks to hold enough high-quality liquid assets (including reserves) to withstand a liquidity stress scenario for 30 days, and the liquidity requirements included in the Resolution Plans introduced by Regulation QQ. Horizontal shifts can also be due to changes in banks' internal risk-management and stress-testing models.

Panel (b) of Figure C.1 shows structural time-varying estimates of the slopes of the reserve demand curve in the U.S. over 2010–June 2025 from Afonso et al. (2025). Afonso et al. (2025) use these estimates to detect horizontal shifts and identify the drivers of vertical shifts; they find that (i) the curve is flat for reserves above a satiation level, as predicted by the theory and consistent with the demand curve in equation (2); (ii) the satiation point moved horizontally to the right by one percentage point of bank assets, from \$1.5 trillion of reserves in 2011 to \$2.2 trillion in 2018, consistent with a horizontal shift to the demand curve captured by the shock ϵ in our model; (iii) a temporary regulatory change in the SLR calculation shifted the curve upward by roughly 5 basis points in 2020–2021, consistent with the evidence in Panel (a) and represented in our model by the vertical shock ν .²¹

²¹In April 2020, to ease strains in the Treasury market resulting from the Covid pandemic, U.S. regulators announced revisions to temporarily exclude reserves (and Treasuries) from the SLR calculation, reducing the regulatory cost of borrowing in the federal funds market. The relief expired on March 31, 2021.

Figure C.1: **Demand for Reserves.** Panel (a): The figure is taken from Afonso et al. (2025) and plots the relationship between the federal funds spread and aggregate reserves from January 2010 to June 2025. The spread is the difference between the daily volume-weighted average federal funds rate and the interest rate on reserve balances (IORB). Reserves are normalized relative to commercial banks' assets. One-day windows around month-ends are excluded to control for the window-dressing of European banks in the federal funds market. Panel (b): The figure plots the structural time-varying daily estimates of the slope of the reserve demand curve from Afonso et al. (2025).

(a) "Realized" Reserve Demand Curve



(b) Structural Time-varying Estimates of the Slope

